9702/22/M/J/09/Q2

1 An experiment is conducted on the surface of the planet Mars.
A sphere of mass 0.78 kg is projected almost vertically upwards from the surface of the planet. The variation with time t of the vertical velocity v in the upward direction is shown in Fig. 2.1.

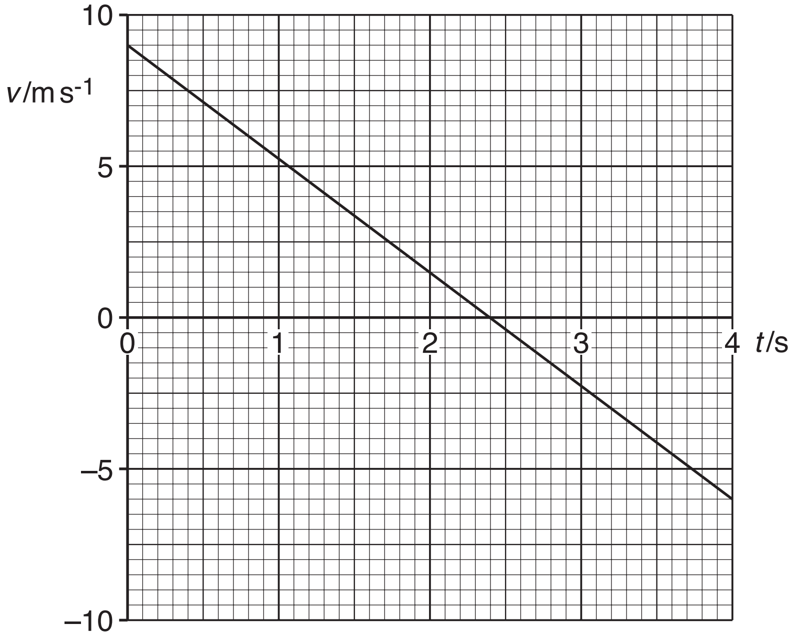


Fig. 2.1

The sphere lands on a small hill at time t = 4.0 s.

(a) State the time t at which the sphere reaches its maximum height above the planet's surface.

t = ... s [1]

(b) Determine the vertical height above the point of projection at which the sphere finally comes to rest on the hill.

height = ... m [3]

(c) Calculate, for the first 3.5 s of the motion of the sphere,

(i) the change in momentum of the sphere,

change in momentum = ... N s [2]

(ii) the force acting on the sphere.

force = ... N [2]

(d) Using your answer in **(c)(ii)**,

(i) state the weight of the sphere,

weight = ... N [1]

(ii) determine the acceleration of free fall on the surface of Mars.

acceleration = ... $m s^{-2}$ [2]

9702/21/O/N/09/Q3

2 A stationary nucleus of mass 220u undergoes radioactive decay to produce a nucleus D of mass 216u and an α-particle of mass 4u, as illustrated in Fig. 3.1.

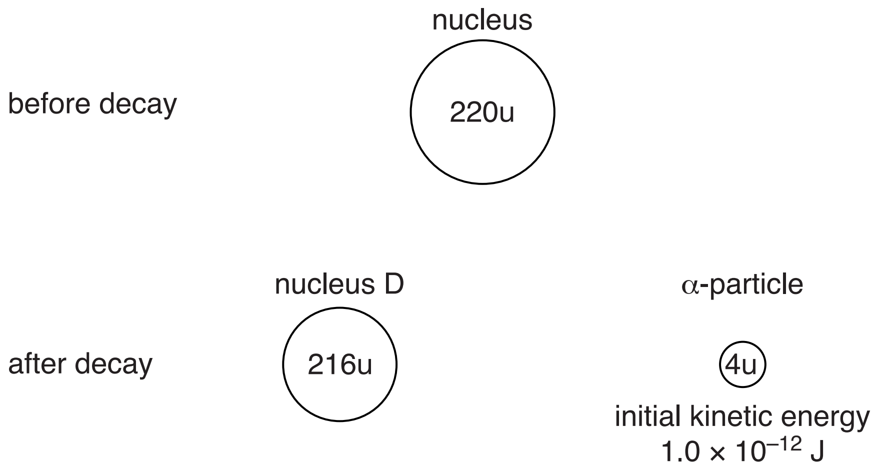


Fig. 3.1

The initial kinetic energy of the α-particle is 1.0×10^{-12} J.

(a) **(i)** State the law of conservation of linear momentum.

....

....

.... [2]

(ii) Explain why the initial velocities of the nucleus D and the α-particle must be in opposite directions.

....

....

.... [2]

(b) **(i)** Show that the initial speed of the α-particle is $1.7 \times 10^{7}\,\mathrm{m\,s^{-1}}$.

[2]

(ii) Calculate the initial speed of nucleus D.

speed = ... $\mathrm{m\,s^{-1}}$ [2]

(c) The range in air of the emitted α-particle is 4.5 cm.
Calculate the average deceleration of the α-particle as it is stopped by the air.

deceleration = ... $\mathrm{m\,s^{-2}}$ [2]

9702/22/M/J/10/Q3

3 **(a)** **(i)** Define *force*.

....

....[1]

(ii) State Newton's third law of motion.

....

....

....

....[3]

(b) Two spheres approach one another along a line joining their centres, as illustrated in Fig. 3.1.

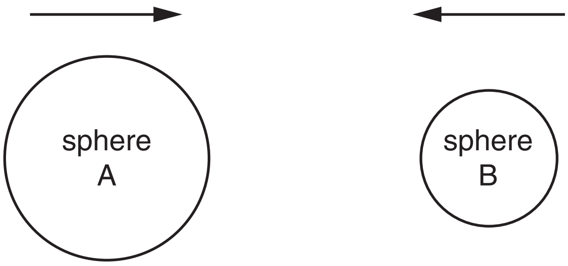


Fig. 3.1

When they collide, the average force acting on sphere A is F_A and the average force acting on sphere B is F_B.

The forces act for time t_A on sphere A and time t_B on sphere B.

(i) State the relationship between

1. F_A and F_B,

....[1]

2. t_A and t_B.

....[1]

(ii) Use your answers in **(i)** to show that the change in momentum of sphere A is equal in magnitude and opposite in direction to the change in momentum of sphere B.

....

....[1]

9702/21/O/N/10/Q3

4 (a) State the relation between force and momentum.

.... [1]

(b) A rigid bar of mass 450 g is held horizontally by two supports A and B, as shown in Fig. 3.1.

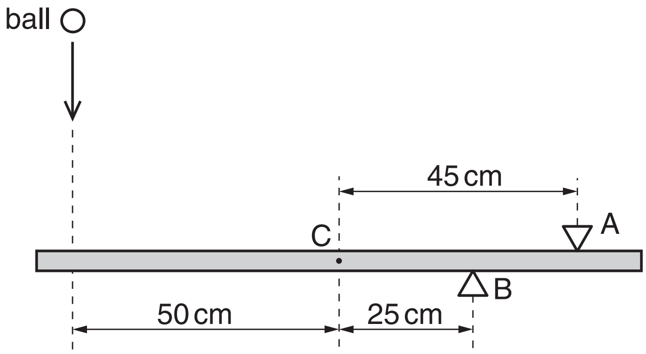


Fig. 3.1

The support A is 45 cm from the centre of gravity C of the bar and support B is 25 cm from C.

A ball of mass 140 g falls vertically onto the bar such that it hits the bar at a distance of 50 cm from C, as shown in Fig. 3.1.
The variation with time t of the velocity v of the ball before, during and after hitting the bar is shown in Fig. 3.2.

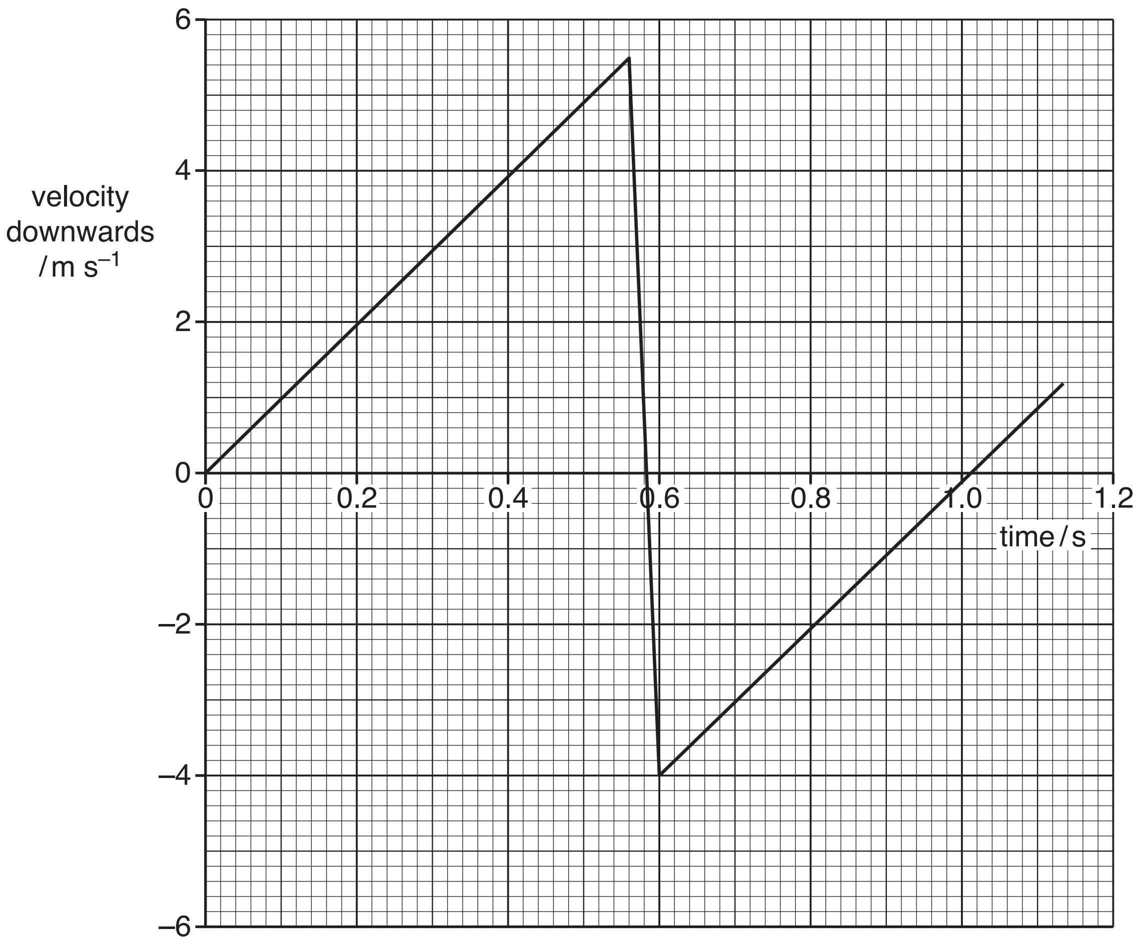


Fig. 3.2

For the time that the ball is in contact with the bar, use Fig. 3.2

(i) to determine the change in momentum of the ball,

change = ... $kg\,m\,s^{-1}$ [2]

(ii) to show that the force exerted by the ball on the bar is 33 N.

[1]

(c) For the time that the ball is in contact with the bar, use data from Fig. 3.1 and **(b)(ii)** to calculate the force exerted on the bar by

(i) the support A,

force = ... N [3]

(ii) the support B.

force = ... N [2]

5 (a) State Newton's second law. **9702/22/O/N/12/Q2**

....

.... [1]

(b) A ball of mass 65 g hits a wall with a velocity of $5.2\,\mathrm{m\,s^{-1}}$ perpendicular to the wall. The ball rebounds perpendicularly from the wall with a speed of $3.7\,\mathrm{m\,s^{-1}}$. The contact time of the ball with the wall is 7.5 ms.

Calculate, for the ball hitting the wall,

(i) the change in momentum,

change in momentum = ... N s [2]

(ii) the magnitude of the average force.

force = ... N [1]

(c) (i) For the collision in **(b)** between the ball and the wall, state how the following apply:

1. Newton's third law,

....

....

....

.... [2]

2. the law of conservation of momentum.

....

.... [1]

(ii) State, with a reason, whether the collision is elastic or inelastic.

....

.... [1]

9702/22/M/J/13/Q2

6 **(a)** Define *force*.

.... [1]

(b) A resultant force F acts on an object of mass 2.4 kg. The variation with time t of F is shown in Fig. 2.1.

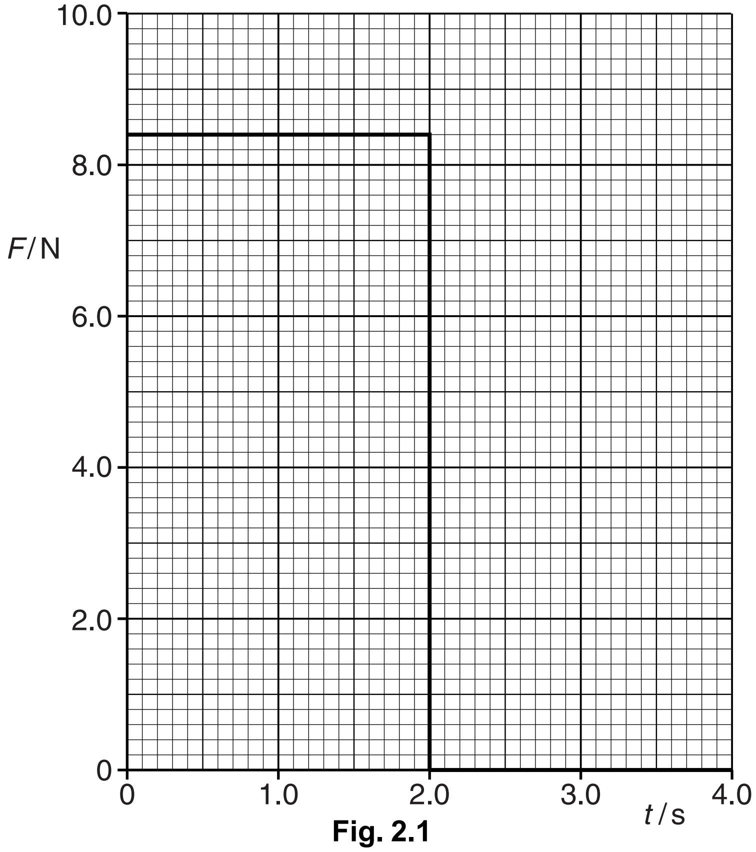
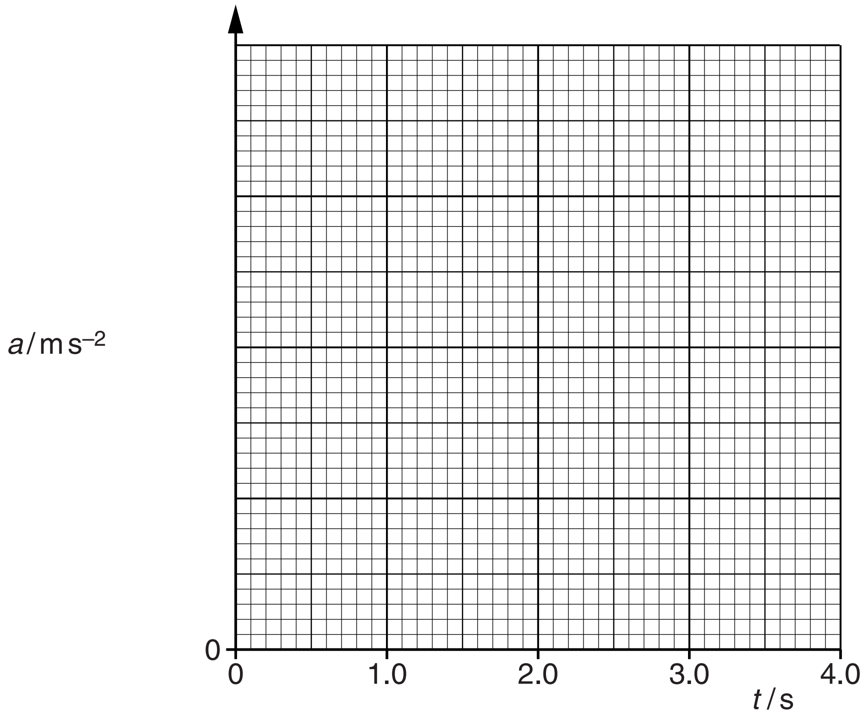


Fig. 2.1

The object starts from rest.

(i) On Fig. 2.2, show quantitatively the variation with t of the acceleration a of the object. Include appropriate values on the y-axis.



[4]

Fig. 2.2

(ii) On Fig. 2.3, show quantitatively the variation with t of the momentum p of the object. Include appropriate values on the y-axis.

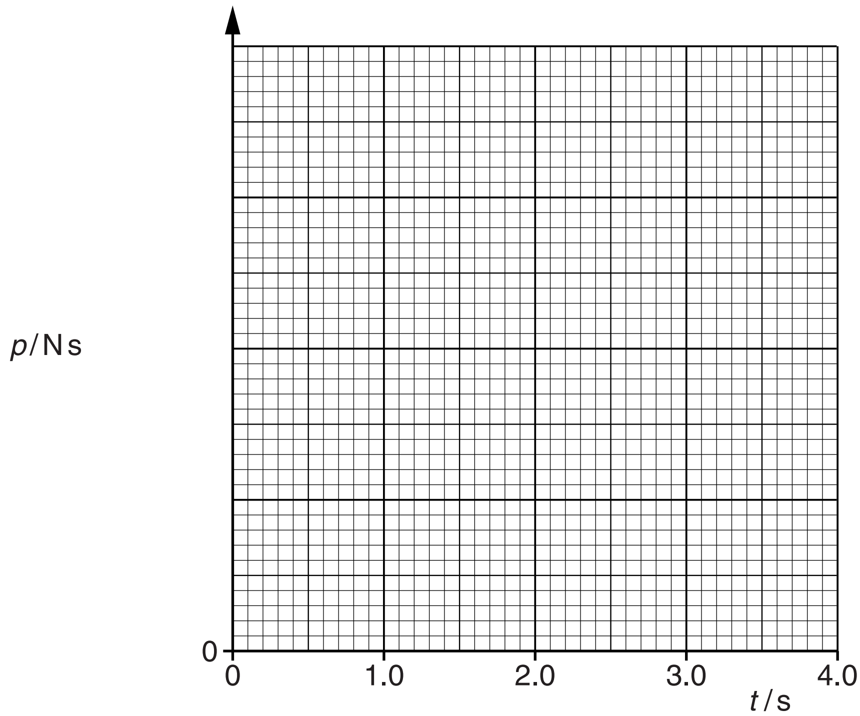


Fig. 2.3

[5]

9702/23/M/J/13/Q3

7 **(a)** **(i)** State the principle of conservation of momentum.

....

....

.... [2]

(ii) State the difference between an elastic and an inelastic collision.

.... [1]

(b) An object A of mass 4.2 kg and horizontal velocity 3.6 m s^{-1} moves towards object B as shown in Fig. 3.1.

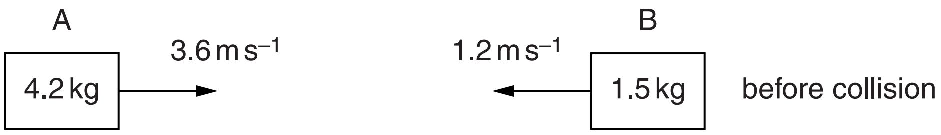


Fig. 3.1

Object B of mass 1.5 kg is moving with a horizontal velocity of 1.2 m s^{-1} towards object A.

The objects collide and then both move to the right, as shown in Fig. 3.2.

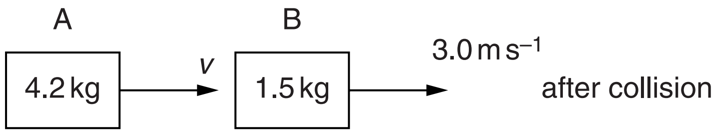


Fig. 3.2

Object A has velocity v and object B has velocity 3.0 m s^{-1}.

(i) Calculate the velocity v of object A after the collision.

velocity = ... m s^{-1} [3]

(ii) Determine whether the collision is elastic or inelastic.

[3]

9702/22/M/J/14/Q3

8 (a) State Newton's first law of motion.

....

.... [1]

(b) A box slides down a slope, as shown in Fig. 3.1.

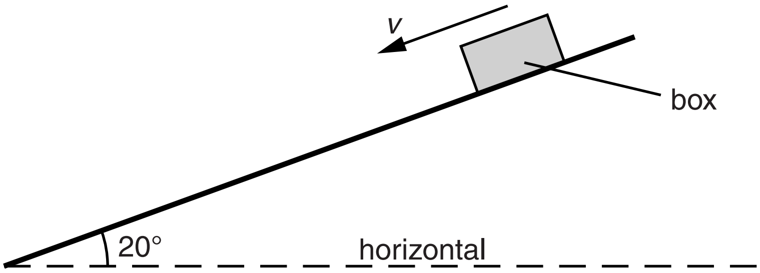


Fig. 3.1

The angle of the slope to the horizontal is 20°. The box has a mass of 65 kg. The total resistive force *R* acting on the box is constant as it slides down the slope.

(i) State the names and directions of the other two forces acting on the box.

1. ....

2. ....
[2]

(ii) The variation with time *t* of the velocity *v* of the box as it moves down the slope is shown in Fig. 3.2.

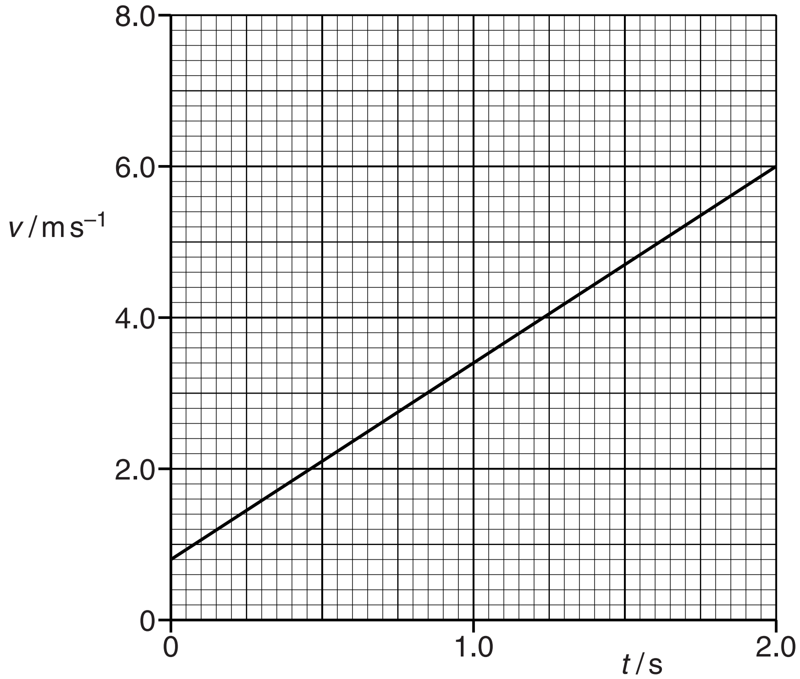


Fig. 3.2

1. Use data from Fig. 3.2 to show that the acceleration of the box is 2.6 $m s^{-2}$.

[2]

2. Calculate the resultant force on the box.

resultant force = ... N [1]

3. Determine the resistive force *R* on the box.

R = ... N [3]

9702/23/O/N/14/Q4

9 **(a)** State the principle of conservation of momentum.

....

....

.... [2]

(b) A ball X and a ball Y are travelling along the same straight line in the same direction, as shown in Fig. 4.1.

Fig. 4.1

Ball X has mass 400 g and horizontal velocity 0.65 m s^{-1}.
Ball Y has mass 600 g and horizontal velocity 0.45 m s^{-1}.

Ball X catches up and collides with ball Y. After the collision, X has horizontal velocity 0.41 m s^{-1} and Y has horizontal velocity v, as shown in Fig. 4.2.

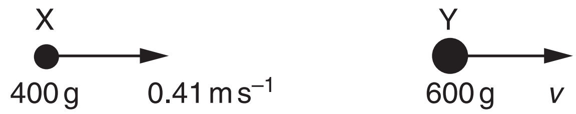


Fig. 4.2

Calculate

(i) the total initial momentum of the two balls,

momentum = ... N s [3]

(ii) the velocity v,

v = ... m s^{-1} [2]

(iii) the total initial kinetic energy of the two balls.

kinetic energy = ... J [3]

(c) Explain how you would check whether the collision is elastic.

....

.... [1]

(d) Use Newton's third law to explain why, during the collision, the change in momentum of X is equal and opposite to the change in momentum of Y.

....

....

.... [2]

9702/21/M/J/15/Q3

10 Two balls X and Y are supported by long strings, as shown in Fig. 3.1.

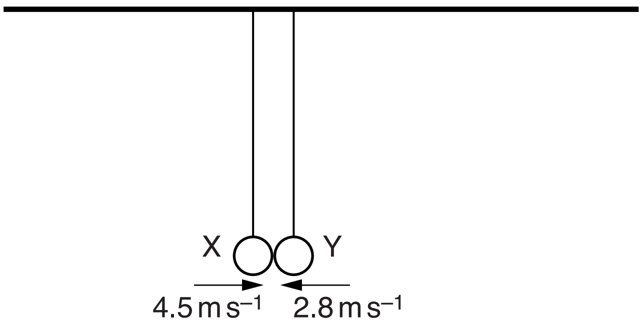


Fig. 3.1

The balls are each pulled back and pushed towards each other. When the balls collide at the position shown in Fig. 3.1, the strings are vertical. The balls rebound in opposite directions.

Fig. 3.2 shows data for X and Y during this collision.

ball	mass	velocity just before collision / m s^{-1}	velocity just after collision / m s^{-1}
X	50 g	+4.5	–1.8
Y	*M*	–2.8	+1.4

Fig. 3.2

The positive direction is horizontal and to the right.

(a) Use the conservation of linear momentum to determine the mass *M* of Y.

M = ... g [3]

(b) State and explain whether the collision is elastic.

....

....

....[1]

(c) Use Newton's second and third laws to explain why the magnitude of the change in momentum of each ball is the same.

....

....

....

....[3]

9702/22/M/J/15/Q4

11 (a) A gas molecule has a mass of 6.64×10^{-27} kg and a speed of $1250\,\mathrm{m\,s^{-1}}$. The molecule collides normally with a flat surface and rebounds with the same speed, as shown in Fig. 4.1.

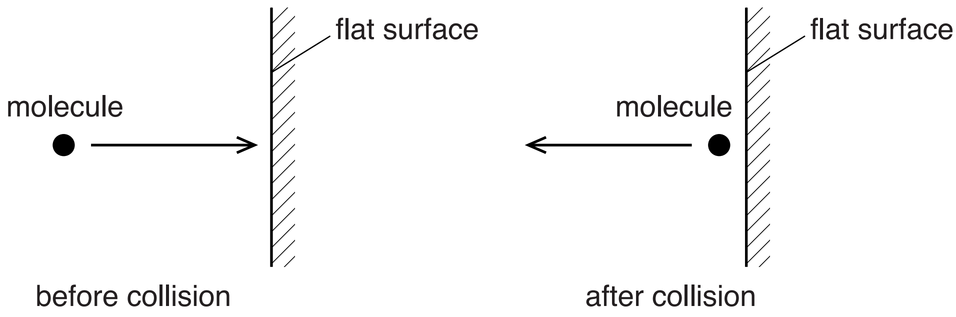


Fig. 4.1

Calculate the change in momentum of the molecule.

change in momentum = ... N s [2]

(b) (i) Use the kinetic model to explain the pressure exerted by gases.

....

....

....

....

....

.... [3]

(ii) Explain the effect of an increase in density, at constant temperature, on the pressure of a gas.

....

.... [1]

(c) For the spheres in **(b)**, the variation with time of the momentum of sphere A before during and after the collision with sphere B is shown in Fig. 3.2.

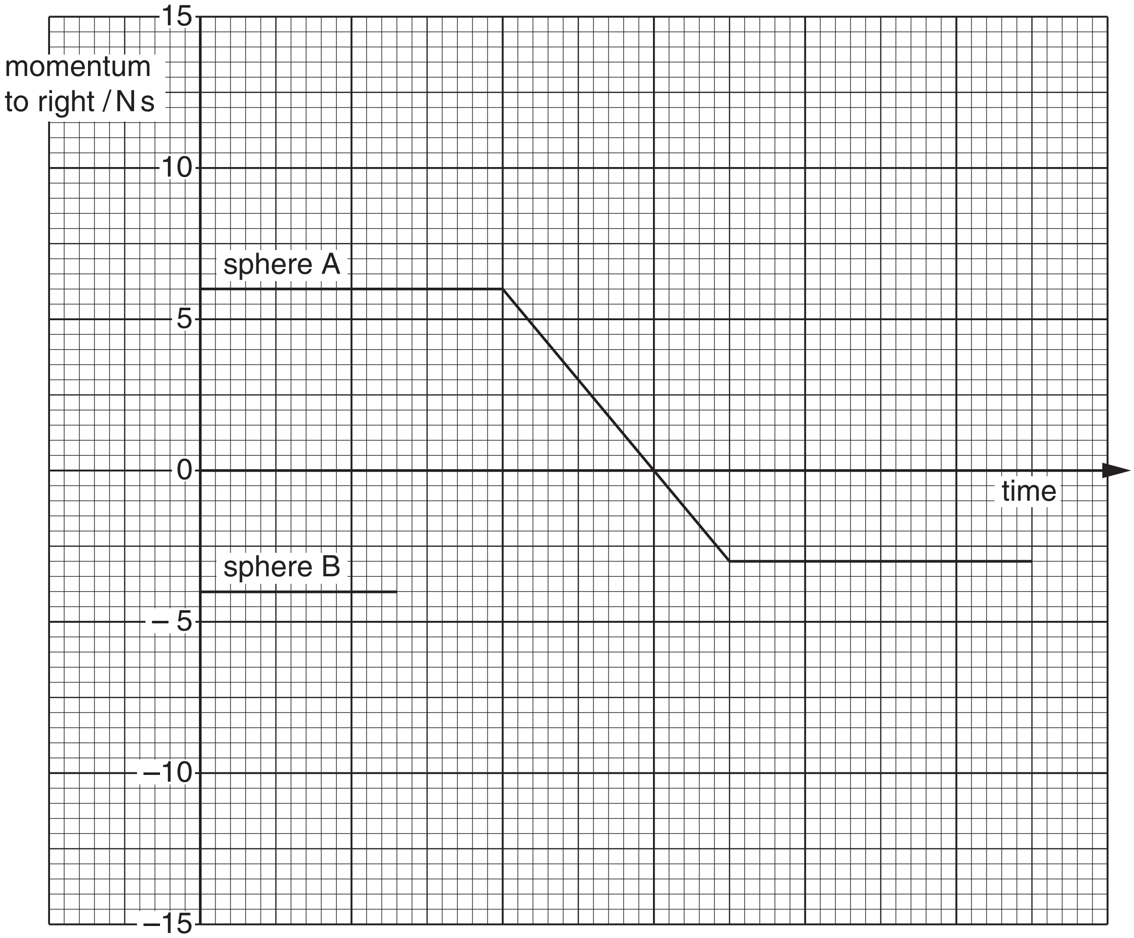


Fig. 3.2

The momentum of sphere B before the collision is also shown on Fig. 3.2.

Complete Fig. 3.2 to show the variation with time of the momentum of sphere B during and after the collision with sphere A.

[3]

12 A ball of mass 150 g is at rest on a horizontal floor, as shown in Fig. 3.1. **9702/21/M/J/16/Q3**

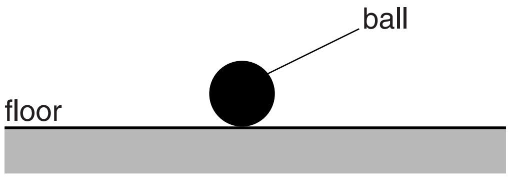


Fig. 3.1

(a) **(i)** Calculate the magnitude of the normal contact force from the floor acting on the ball.

force = ... N [1]

(ii) Explain your working in **(i)**.

....

....

....[1]

(b) The ball is now lifted above the floor and dropped so that it falls vertically, as illustrated in Fig. 3.2.

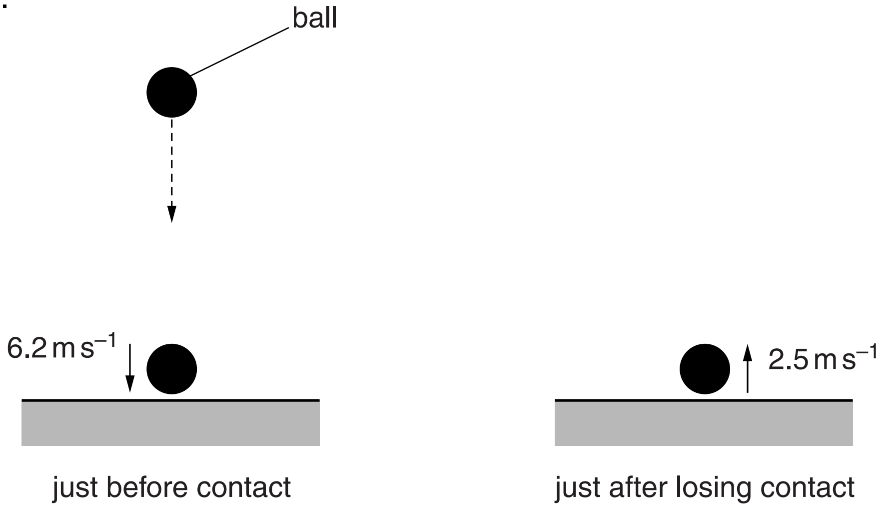


Fig. 3.2

Just before contact with the floor, the ball has velocity 6.2 m s^{-1} downwards. The ball bounces from the floor and its velocity just after losing contact with the floor is 2.5 m s^{-1} upwards. The ball is in contact with the floor for 0.12 s.

(i) State Newton's second law of motion.

....

....[1]

(ii) Calculate the average resultant force on the ball when it is in contact with the floor.

magnitude of force = ... N

direction of force ....

[3]

(iii) State and explain whether linear momentum is conserved during the collision of the ball with the floor.

....

....

....

....[2]

13 (a) State the law of conservation of momentum. **9702/23/M/J/16/Q5**

....

....

.... [2]

(b) Two particles A and B collide elastically, as illustrated in Fig. 5.1.

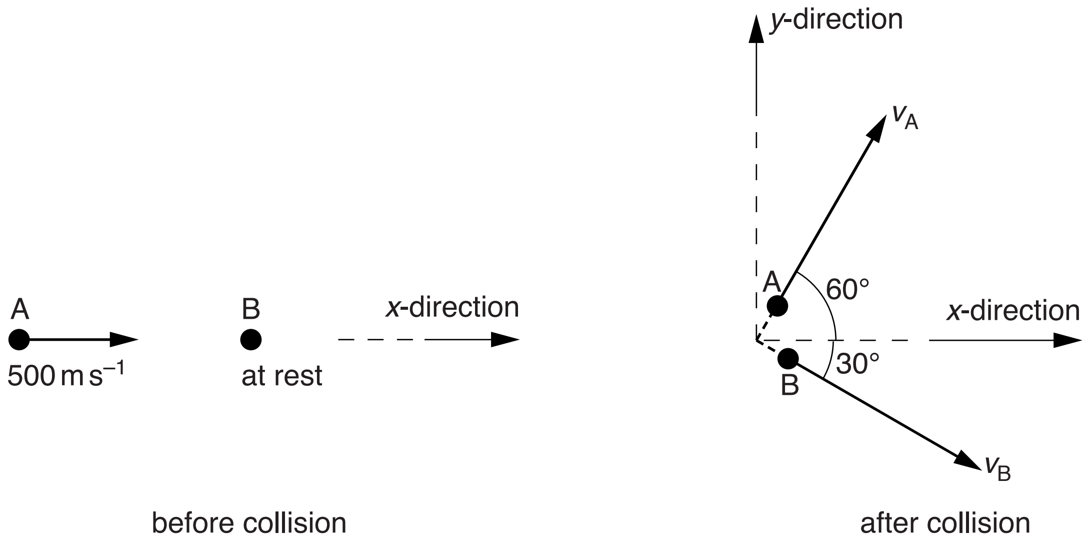


Fig. 5.1

The initial velocity of A is 500 m s^{-1} in the x-direction and B is at rest.

The velocity of A after the collision is v_A at 60° to the x-direction. The velocity of B after the collision is v_B at 30° to the x-direction.

The mass m of each particle is 1.67×10^{-27} kg.

(i) Explain what is meant by the particles colliding *elastically*.

.... [1]

(ii) Calculate the total initial momentum of A and B.

momentum = ... N s [1]

(iii) State an expression in terms of m, v_A and v_B for the total momentum of A and B after the collision

1. in the x-direction,

....

2. in the y-direction.

....
[2]

(iv) Calculate the magnitudes of the velocities v_A and v_B after the collision.

v_A = ... $m\,s^{-1}$

v_B = ... $m\,s^{-1}$
[3]

9702/21/M/J/17/Q2 (b)

14 A paraglider P of mass 95 kg is pulled by a wire attached to a boat, as shown in Fig. 2.1.

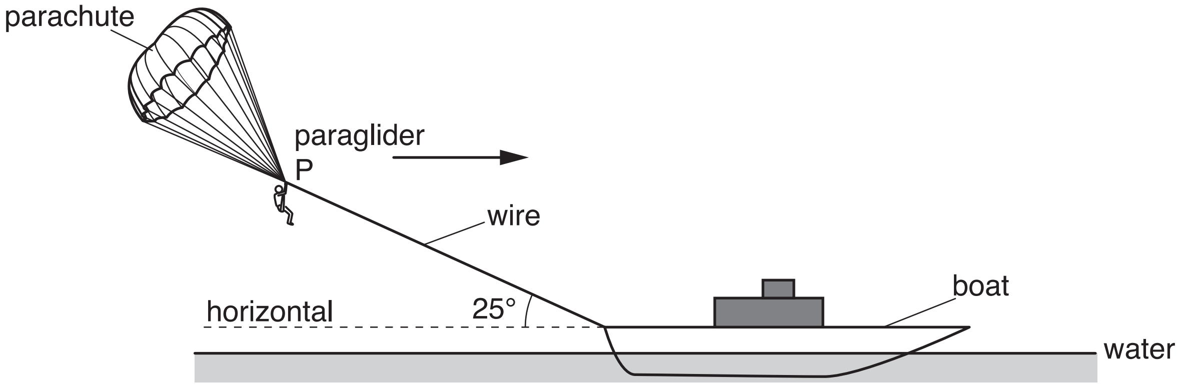


Fig. 2.1

The wire makes an angle of 25° with the horizontal water surface. P moves in a straight line parallel to the surface of the water.

The variation with time t of the velocity v of P is shown in Fig. 2.2.

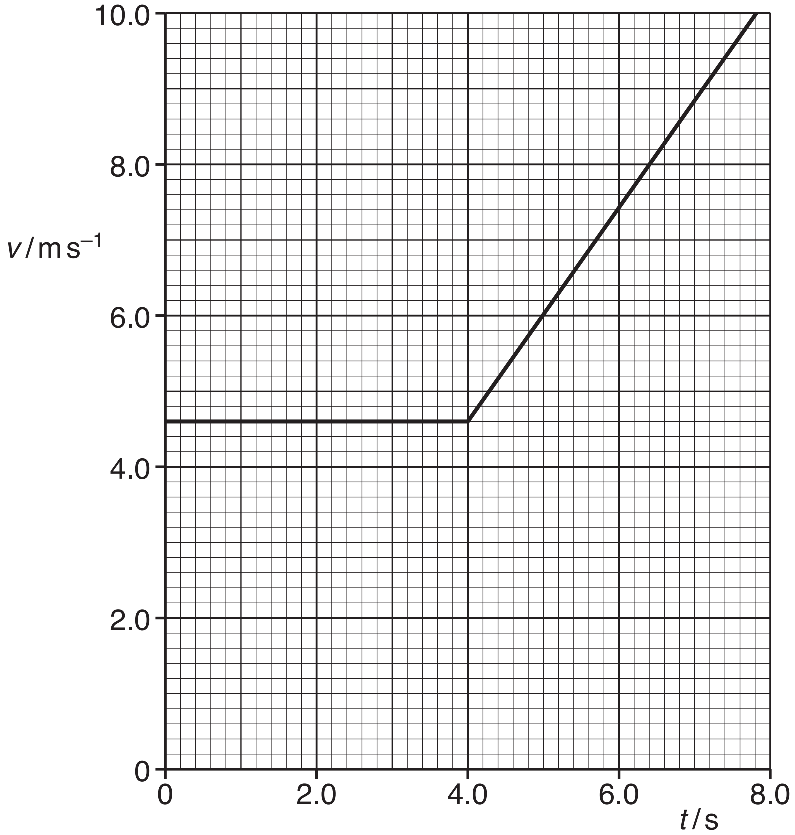


Fig. 2.2

(i) Show that the acceleration of P is 1.4 $m\,s^{-2}$ at time $t = 5.0\,s$.

[2]

(ii) Calculate the total distance moved by P from time $t = 0$ to $t = 7.0\,s$.

distance = ... m [2]

(iii) Calculate the change in kinetic energy of P from time $t = 0$ to $t = 7.0\,s$.

change in kinetic energy = ... J [2]

(iv) The tension in the wire at time $t = 5.0\,s$ is 280 N.

Calculate, for the horizontal motion,

1. the vertical lift force F supporting P,

F = ... N [3]

2. the force R due to air resistance acting on P in the horizontal direction.

R = ... N [3]

9702/22/M/J/17/Q4

15 **(a)** State Newton's first law of motion.

....

.... [1]

(b) An object A of mass 100 g is moving in a straight line with a velocity of $0.60\,\mathrm{m\,s^{-1}}$ to the right. An object B of mass 200 g is moving in the same straight line as object A with a velocity of $0.80\,\mathrm{m\,s^{-1}}$ to the left, as shown in Fig. 4.1.

Fig. 4.1

Objects A and B collide. Object A then moves with a velocity of $0.40\,\mathrm{m\,s^{-1}}$ to the left.

(i) Calculate the magnitude of the velocity of B after the collision.

magnitude of velocity = ... $\mathrm{m\,s^{-1}}$ [2]

(ii) The collision between A and B is inelastic.

Explain how the collision is inelastic and still obeys the law of conservation of energy.

....

....

.... [1]

9702/21/O/N/17/Q1 (b)

16 (a) A ball of weight 1.5 N falls vertically from rest in air. The drag force F_D acting on the ball is given by the expression in **(a)**. The ball reaches a constant (terminal) speed of 33 m s^{-1}.

Assume that the upthrust acting on the ball is negligible and that the density of the air is uniform.

For the instant when the ball is travelling at a speed of 25 m s^{-1}, determine

(i) the drag force F_D on the ball,

F_D = ... N [2]

(ii) the acceleration of the ball.

acceleration = ... m s^{-2} [2]

(b) Describe the acceleration of the ball in **(b)** as its speed changes from zero to 33 m s^{-1}.

....

....

....

....[3]

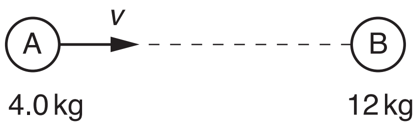
17 **(a)** State the principle of conservation of momentum. **9702/23/O/N/17/Q3**

....

....

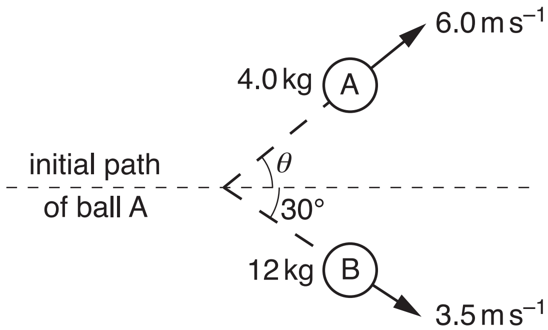
....[2]

(b) Ball A moves with speed *v* along a horizontal frictionless surface towards a stationary ball B, as shown in Fig. 3.1.



before collision

Fig. 3.1



after collision

Fig. 3.2 (not to scale)

Ball A has mass 4.0 kg and ball B has mass 12 kg.
The balls collide and then move apart as shown in Fig. 3.2.
Ball A has velocity 6.0 m s^{-1} at an angle of θ to the direction of its initial path.
Ball B has velocity 3.5 m s^{-1} at an angle of 30° to the direction of the initial path of ball A.

(i) By considering the components of momentum at right-angles to the direction of the initial path of ball A, calculate θ.

θ = ... ° [3]

(ii) Use your answer in **(i)** to show that the initial speed *v* of ball A is 12 m s^{-1}.
Explain your working.

[2]

(iii) By calculation of kinetic energies, state and explain whether the collision is elastic or inelastic.

....

....[3]

9702/21/M/J/18/Q3

18 **(a)** State what is meant by the *mass* of a body.

....

....[1]

(b) Two blocks travel directly towards each other along a horizontal, frictionless surface. The blocks collide, as illustrated in Fig. 3.1.

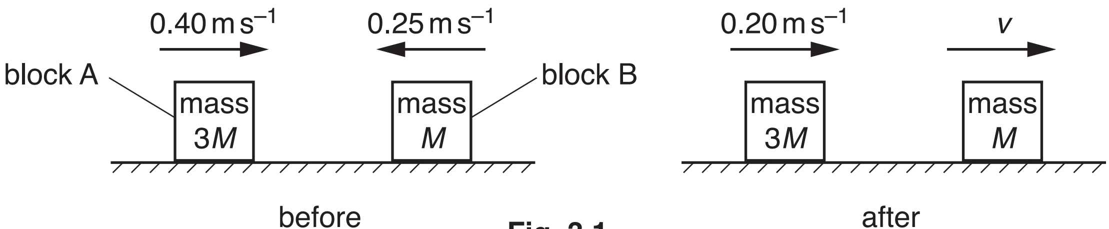


Fig. 3.1

Block A has mass $3M$ and block B has mass M.
Before the collision, block A moves to the right with speed $0.40\,\mathrm{m\,s^{-1}}$ and block B moves to the left with speed $0.25\,\mathrm{m\,s^{-1}}$.
After the collision, block A moves to the right with speed $0.20\,\mathrm{m\,s^{-1}}$ and block B moves to the right with speed v.

(i) Use Newton's third law to explain why, during the collision, the change in momentum of block A is equal and opposite to the change in momentum of block B.

....

....

....[2]

(ii) Determine speed v.

v = ... $m\,s^{-1}$ [3]

(iii) Calculate, for the blocks,

1. the relative speed of approach,

relative speed of approach = ... $m\,s^{-1}$

2. the relative speed of separation.

relative speed of separation = ... $m\,s^{-1}$ [2]

(iv) Use your answers in **(b)(iii)** to state and explain whether the collision is elastic or inelastic.

....

....[1]

9702/22/M/J/18/Q2

19 **(a)** State the principle of conservation of momentum.

....

....

....[2]

(b) A stationary firework explodes into three different fragments that move in a horizontal plane, as illustrated in Fig. 2.1.

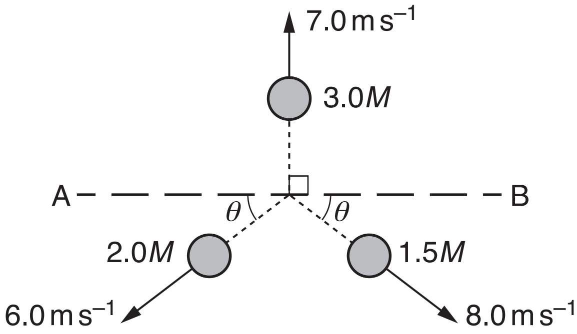


Fig. 2.1

The fragment of mass 3.0M has a velocity of 7.0 m s^{-1} perpendicular to line AB.
The fragment of mass 2.0M has a velocity of 6.0 m s^{-1} at angle θ to line AB.
The fragment of mass 1.5M has a velocity of 8.0 m s^{-1} at angle θ to line AB.

(i) Use the principle of conservation of momentum to determine θ.

θ = ... ° [3]

(ii) Calculate the ratio

$$\frac{\text{kinetic energy of fragment of mass } 2.0M}{\text{kinetic energy of fragment of mass } 1.5M}.$$

ratio = ...[2]

9702/21/O/N/18/Q2

20 A wooden block moves along a horizontal frictionless surface, as shown in Fig. 2.1.

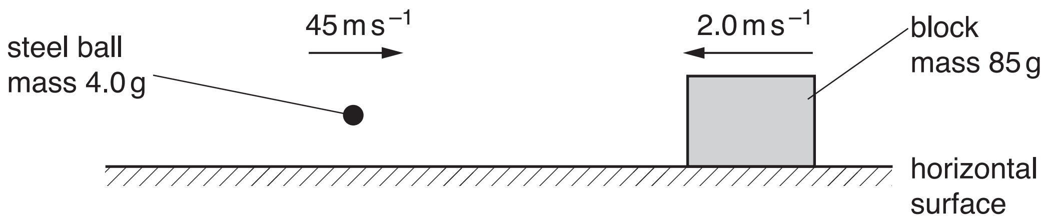


Fig. 2.1

The block has mass 85 g and moves to the left with a velocity of 2.0 m s^{-1}. A steel ball of mass 4.0 g is fired to the right. The steel ball, moving horizontally with a speed of 45 m s^{-1}, collides with the block and remains embedded in it. After the collision the block and steel ball both have speed *v*.

(a) Calculate *v*.

v = ... m s^{-1} [2]

(b) (i) For the block and ball, state

1. the relative speed of approach before collision,

relative speed of approach = ... m s^{-1}

2. the relative speed of separation after collision.

relative speed of separation = ... m s^{-1}
[1]

(ii) Use your answers in **(i)** to state and explain whether the collision is elastic or inelastic.

....

.... [1]

(c) Use Newton's third law to explain the relationship between the rate of change of momentum of the ball and the rate of change of momentum of the block during the collision.

....

....

....

.... [2]

9702/22/O/N/18/Q3

21 (a) State the principle of conservation of momentum.

....

....

....[2]

(b) The propulsion system of a toy car consists of a propeller attached to an electric motor, as illustrated in Fig. 3.1.

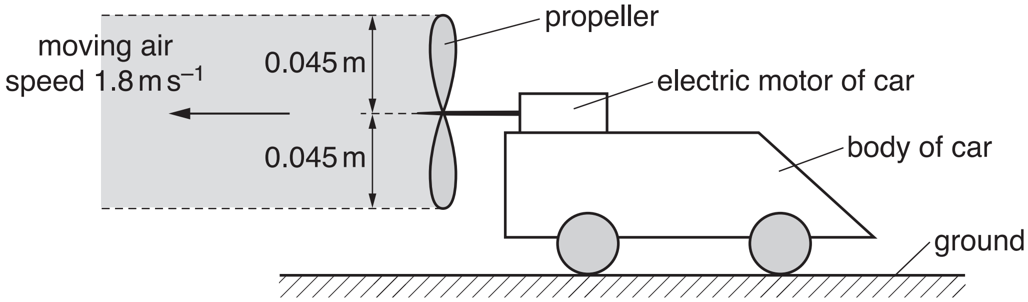


Fig. 3.1

The car is on horizontal ground and is initially held at rest by its brakes. When the motor is switched on, it rotates the propeller so that air is propelled horizontally to the left. The density of the air is 1.3 kg m^{-3}.

Assume that the air moves with a speed of 1.8 m s^{-1} in a uniform cylinder of radius 0.045 m. Also assume that the air to the right of the propeller is stationary.

(i) Show that, in a time interval of 2.0 s, the mass of air propelled to the left is 0.030 kg.

[2]

(ii) Calculate

1. the increase in the momentum of the mass of air in **(b)(i)**,

increase in momentum = ... Ns

2. the force exerted on this mass of air by the propeller.

force = ... N
[3]

(iii) Explain how Newton's third law applies to the movement of the air by the propeller.

....

....

....[2]

(iv) The total mass of the car is 0.20 kg. The brakes of the car are released and the car begins to move with an initial acceleration of $0.075\,\mathrm{m\,s^{-2}}$.

Determine the initial frictional force acting on the car.

frictional force = ... N [2]

9702/23/O/N/18/Q3

22 **(a)** State Newton's second law of motion.

....

....[1]

(b) A toy rocket consists of a container of water and compressed air, as shown in Fig. 3.1.

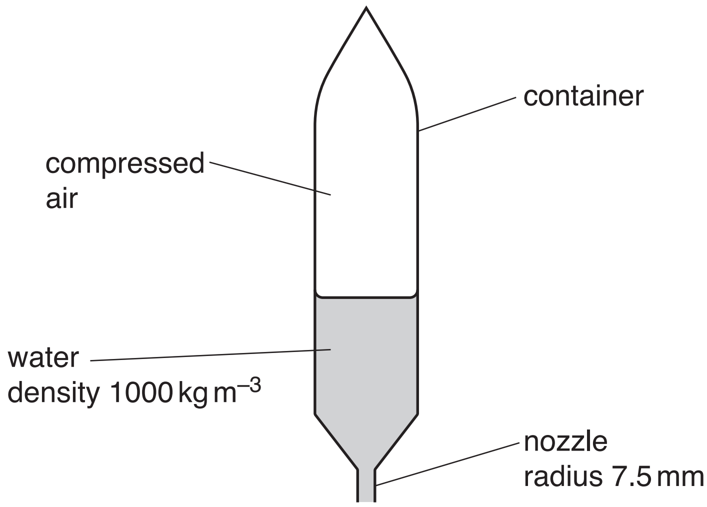


Fig. 3.1

Water is pushed vertically downwards through a nozzle by the compressed air. The rocket moves vertically upwards.

The nozzle has a circular cross-section of radius 7.5 mm. The density of the water is 1000 kg m^{-3}. Assume that the water leaving the nozzle has the shape of a cylinder of radius 7.5 mm and has a constant speed of 13 m s^{-1} relative to the rocket.

(i) Show that the mass of water leaving the nozzle in the first 0.20 s after the rocket launch is 0.46 kg.

[2]

(ii) Calculate

1. the change in the momentum of the mass of water in **(b)(i)** due to leaving the nozzle,

change in momentum = ... N s

2. the force exerted on this mass of water by the rocket.

force = ... N
[3]

(iii) State and explain how Newton's third law applies to the movement of the rocket by the water.

....

....

....[2]

(iv) The container has a mass of 0.40 kg. The initial mass of water before the rocket is launched is 0.70 kg. The mass of the compressed air in the rocket is negligible. Assume that the resistive force on the rocket due to its motion is negligible.

For the rocket at a time of 0.20 s after launching,

1. show that its total mass is 0.64 kg,

2. calculate its acceleration.

acceleration = ... $m\,s^{-2}$
[3]

9702/21/M/J/19/Q2

23 A block X slides along a horizontal frictionless surface towards a stationary block Y, as illustrated in Fig. 2.1.

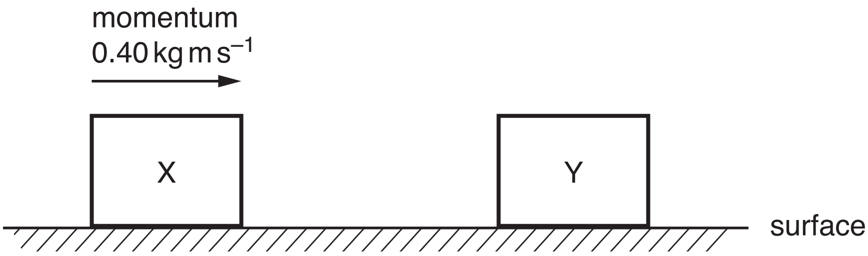


Fig. 2.1

There are no resistive forces acting on block X as it moves towards block Y. At time $t = 0$, block X has momentum 0.40 kg m s^{-1}. A short time later, the blocks collide and then separate.

The variation with time t of the momentum of block Y is shown in Fig. 2.2.

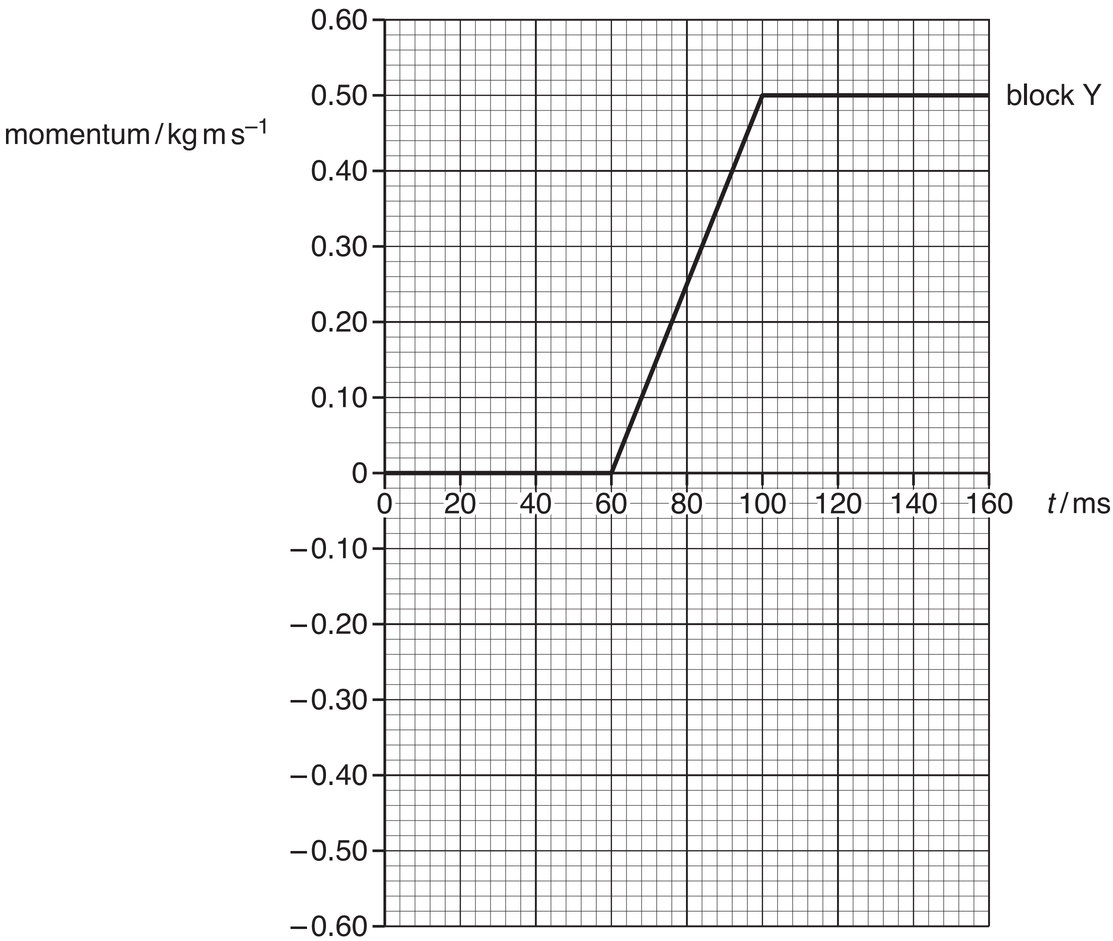


Fig. 2.2

(a) Define *linear momentum*.

....[1]

(b) Use Fig. 2.2 to:

(i) determine the time interval over which the blocks are in contact with each other

time interval = ... ms [1]

(ii) describe, without calculation, the magnitude of the acceleration of block Y from:

1. time t = 80 ms to t = 100 ms

....

2. time t = 100 ms to t = 120 ms.

....
[2]

(c) Use Fig. 2.2 to determine the magnitude of the force exerted by block X on block Y.

force = ... N [2]

(d) On Fig. 2.2, sketch the variation of the momentum of block X with time t from t = 0 to t = 160 ms. [3]

9702/21/M/J/20/Q2

24 **(a)** State Newton's second law of motion.

....

.... [1]

(b) A delivery company suggests using a remote-controlled aircraft to drop a parcel into the garden of a customer. When the aircraft is vertically above point P on the ground, it releases the parcel with a velocity that is horizontal and of magnitude $5.4\,\text{m}\,\text{s}^{-1}$. The path of the parcel is shown in Fig. 2.1.

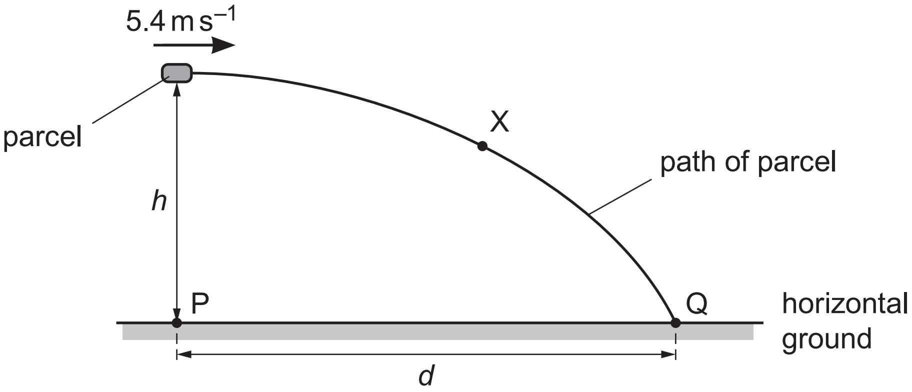


Fig. 2.1 (not to scale)

The parcel takes a time of 0.81 s after its release to reach point Q on the horizontal ground. Assume air resistance is negligible.

(i) On Fig. 2.1, draw an arrow from point X to show the direction of the acceleration of the parcel when it is at that point. [1]

(ii) Determine the height h of the parcel above the ground when it is released.

h = ... m [2]

(iii) Calculate the horizontal distance d between points P and Q.

d = ... m [1]

(c) Another parcel is accidentally released from rest by a different aircraft when it is hovering at a great height above the ground. Air resistance is now significant.

(i) On Fig. 2.2, draw arrows to show the directions of the forces acting on the parcel as it falls vertically downwards. Label each arrow with the name of the force.

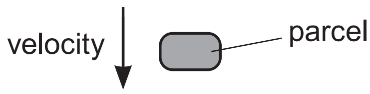


Fig. 2.2

[2]

(ii) By considering the forces acting on the parcel, state and explain the variation, if any, of the acceleration of the parcel as it moves downwards before it reaches constant (terminal) speed.

....

....

....

....

....

.... [3]

(iii) Describe the energy conversion that occurs when the parcel is falling through the air at constant (terminal) speed.

....

.... [1]

9702/22/M/J/20/Q2

25 **(a)** Fig. 2.1 shows the velocity–time graph for an object moving in a straight line.

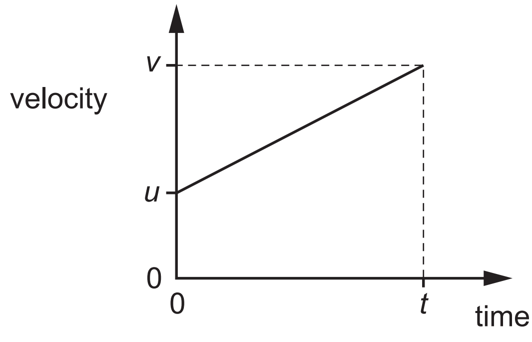


Fig. 2.1

(i) Determine an expression, in terms of u, v and t, for the area under the graph.

area = ... [1]

(ii) State the name of the quantity represented by the area under the graph.

.... [1]

(b) A ball is kicked with a velocity of $15\,\text{m}\,\text{s}^{-1}$ at an angle of 60° to horizontal ground. The ball then strikes a vertical wall at the instant when the path of the ball becomes horizontal, as shown in Fig. 2.2.

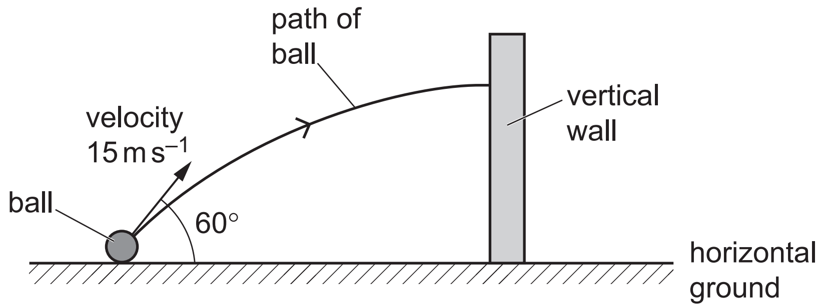


Fig. 2.2 (not to scale)

Assume that air resistance is negligible.

(i) By considering the vertical motion of the ball, calculate the time it takes to reach the wall.

time = ... s [3]

(ii) Explain why the horizontal component of the velocity of the ball remains constant as it moves to the wall.

....

.... [1]

(iii) Show that the ball strikes the wall with a horizontal velocity of $7.5\,\mathrm{m\,s^{-1}}$.

[1]

(c) The mass of the ball in **(b)** is 0.40 kg. It is in contact with the wall for a time of 0.12 s and rebounds horizontally with a speed of $4.3\,\mathrm{m\,s^{-1}}$.

(i) Use the information from **(b)(iii)** to calculate the change in momentum of the ball due to the collision.

change in momentum = ... $\mathrm{kg\,m\,s^{-1}}$ [2]

(ii) Calculate the magnitude of the average force exerted on the ball by the wall.

average force = ... N [1]

9702/23/M/J/20/Q2

26 **(a)** State Newton's first law of motion.

....

.... [1]

(b) A skier is pulled in a straight line along horizontal ground by a wire attached to a kite, as shown in Fig. 2.1.

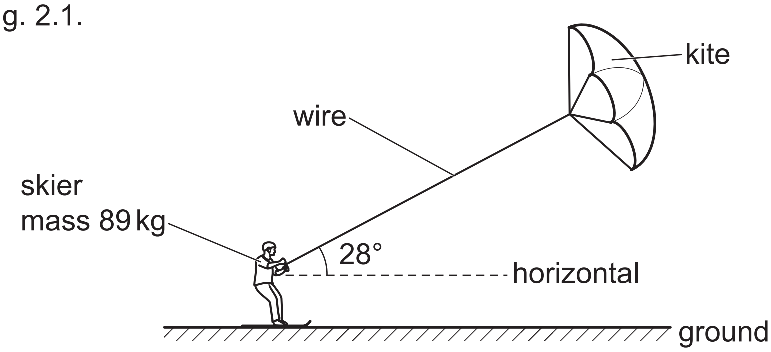


Fig. 2.1 (not to scale)

The mass of the skier is 89 kg. The wire is at an angle of 28° to the horizontal. The variation with time t of the velocity v of the skier is shown in Fig. 2.2.

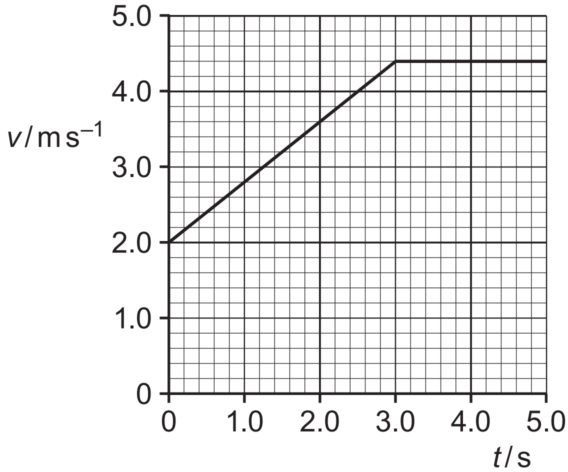


Fig. 2.2

(i) Use Fig. 2.2 to determine the distance moved by the skier from time $t = 0$ to $t = 5.0$ s.

distance = ... m [2]

(ii) Use Fig. 2.2 to show that the acceleration a of the skier is $0.80\,m\,s^{-2}$ at time $t = 2.0\,s$.

[2]

(iii) The tension in the wire at time $t = 2.0\,s$ is 240 N.

Calculate:

1. the horizontal component of the tension force acting on the skier

horizontal component of force = ... N [1]

2. the total resistive force R acting on the skier in the horizontal direction.

R = ... N [2]

(iv) The skier is now lifted upwards by a gust of wind. For a few seconds the skier moves horizontally through the air with the wire at an angle of 45° to the horizontal, as shown in Fig. 2.3.

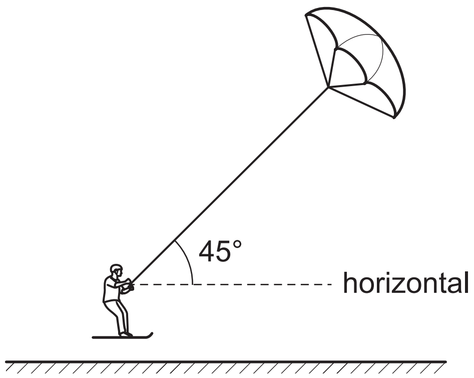


Fig. 2.3 (not to scale)

By considering the vertical components of the forces acting on the skier, determine the new tension in the wire when the skier is moving horizontally through the air.

tension = ... N [2]

27. 9702/21/O/N/20/Q3

(a) Define *force*.

....

.... [1]

(b) A ball falls vertically downwards towards a horizontal floor and then rebounds along its original path, as illustrated in Fig. 3.1.

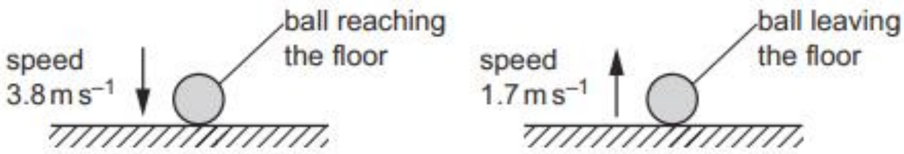


Fig. 3.1

The ball reaches the floor with speed $3.8\,\mathrm{m\,s^{-1}}$. The ball is then in contact with the floor for a time of 0.081 s before leaving it with speed $1.7\,\mathrm{m\,s^{-1}}$. The mass of the ball is 0.062 kg.

(i) Calculate the loss of kinetic energy of the ball during the collision.

loss of kinetic energy = ... J [2]

(ii) Determine the magnitude of the change in momentum of the ball during the collision.

change in momentum = ... N s [2]

(iii) Show that the magnitude of the average resultant force acting on the ball during the collision is 4.2 N.

[1]

(iv) Use the information in **(iii)** to calculate the magnitude of:

1. the average force of the floor on the ball during the collision

average force = ... N

2. the average force of the ball on the floor during the collision.

average force = ... N

[2]

[Total: 8]

28. 9702/22/O/N/20/Q2

(a) A cylinder is suspended from the end of a string. The cylinder is stationary in water with the axis of the cylinder vertical, as shown in Fig. 2.1.

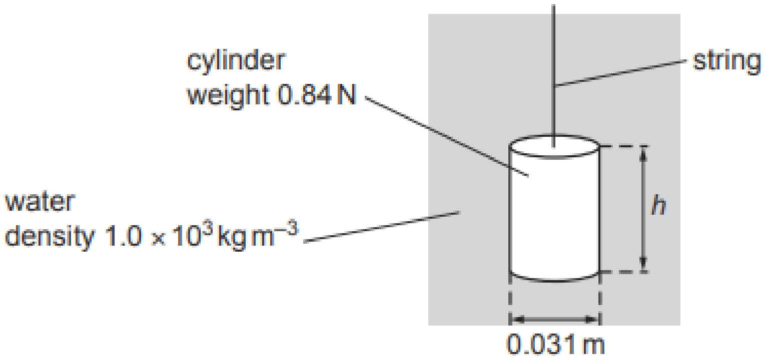


Fig. 2.1 (not to scale)

The cylinder has weight 0.84 N, height h and a circular cross-section of diameter 0.031 m. The density of the water is $1.0 \times 10^3\,kg\,m^{-3}$. The difference between the pressures on the top and bottom faces of the cylinder is 520 Pa.

(i) Calculate the height h of the cylinder.

h = ... m [2]

(ii) Show that the upthrust acting on the cylinder is 0.39 N.

[2]

(iii) Calculate the tension T in the string.

T = ... N [1]

(b) The string is now used to move the cylinder in **(a)** vertically upwards through the water. The variation with time t of the velocity v of the cylinder is shown in Fig. 2.2.

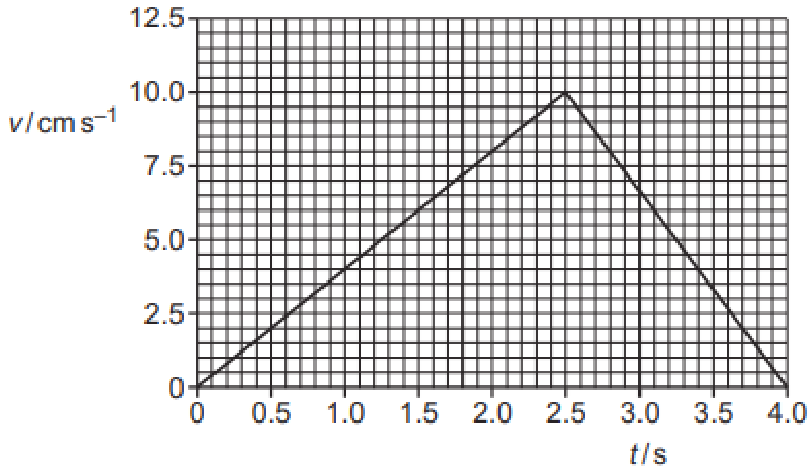


Fig. 2.2

(i) Use Fig. 2.2 to determine the acceleration of the cylinder at time t = 2.0 s.

acceleration = ... m s^{-2} [2]

(ii) The top face of the cylinder is at a depth of 0.32 m below the surface of the water at time t = 0.

Use Fig. 2.2 to determine the depth of the top face below the surface of the water at time t = 4.0 s.

depth = ... m [2]

(c) The cylinder in **(b)** is released from the string at time t = 4.0 s. The cylinder falls, from rest, vertically downwards through the water. Assume that the upthrust acting on the cylinder remains constant as it falls.

(i) State the name of the force that acts on the cylinder when it is moving and does not act on the cylinder when it is stationary.

.... [1]

(ii) State and explain the variation, if any, of the acceleration of the cylinder as it falls downwards through the water.

....

....

.... [2]

[Total: 12]

29. 9702/22/M/J/21/Q1

(a) Complete Table 1.1 by stating whether each of the quantities is a vector or a scalar.

Table 1.1

quantity	vector or scalar
acceleration	
electrical resistance	
momentum	

[2]

(b) State the conditions for an object to be in equilibrium.

....

....

....

.... [2]

(c) A floating solid cylinder is attached by a wire to the sea bed, as shown in Fig. 1.1.

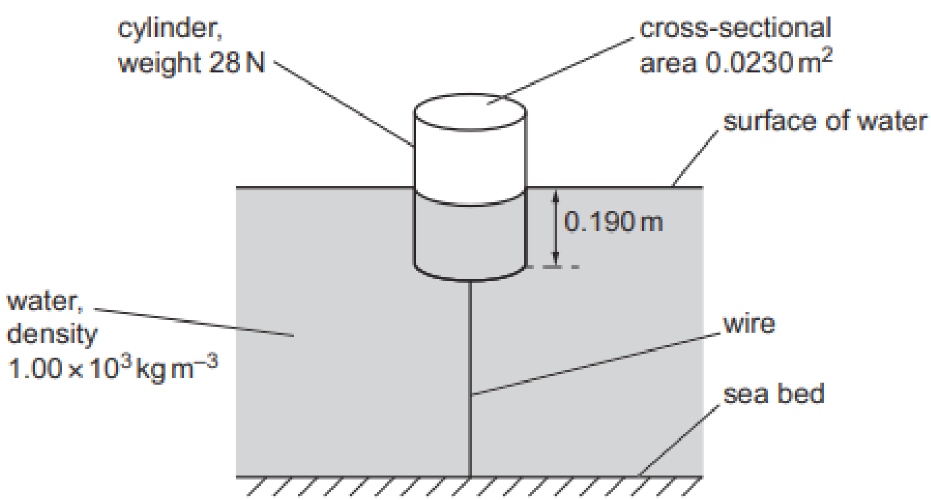


Fig. 1.1 (not to scale)

The density of the water is $1.00 \times 10^3\,\text{kg}\,\text{m}^{-3}$. The base of the cylinder is at a depth of 0.190 m below the surface of the water. The cylinder has a weight of 28 N and a cross-sectional area of $0.0230\,\text{m}^2$.

The wire and the central axis of the cylinder are both vertical. The cylinder is in equilibrium.

(i) Calculate, to three significant figures, the upthrust acting on the cylinder due to the water.

upthrust = ... N [2]

(ii) Show that the tension T in the wire is 15 N.

[1]

(iii) The wire has a cross-sectional area of $3.2\,mm^2$.

Calculate the stress in the wire.

stress = ... Pa [2]

(iv) The surface of the water gradually rises until it is level with the top face of the cylinder.

State and explain, qualitatively, the variation of the strain energy stored in the wire as the water surface rises.

....

....

....

.... [2]

30. 9702/22/M/J/21/Q2

A ball is thrown vertically downwards to the ground, as illustrated in Fig. 2.1.

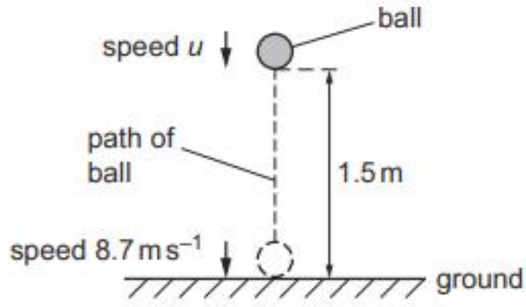


Fig. 2.1

The ball is thrown with speed u from a height of 1.5 m. The ball then hits the ground with speed 8.7 m s^{-1}. Assume that air resistance is negligible.

(a) Calculate speed u.

u = ... m s^{-1} [2]

(b) State how Newton's third law applies to the collision between the ball and the ground.

....

....

....

.... [2]

(c) The ball is in contact with the ground for a time of 0.091 s. The ball rebounds vertically and leaves the ground with speed 5.4 m s^{-1}. The mass of the ball is 0.059 kg.

(i) Calculate the magnitude of the change in momentum of the ball during the collision.

change in momentum = ... N s [2]

(ii) Determine the magnitude of the average resultant force that acts on the ball during the collision.

average resultant force = ... N [1]

(iii) Use your answer in **(c)(ii)** to calculate the magnitude of the average force exerted by the ground on the ball during the collision.

average force = ... N [2]

(d) The ball was thrown downwards at time t = 0 and hits the ground at time t = T.

On Fig. 2.2, sketch a graph to show the variation of the speed of the ball with time t from t = 0 to t = T. Numerical values are not required.

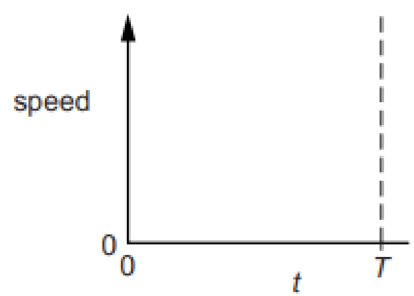


Fig. 2.2

[1]

(e) In practice, air resistance is not negligible.

State and explain the variation, if any, with time t of the gradient of the graph in **(d)** when air resistance is not negligible.

....

....

....

.... [2]

31. 9702/21/O/N/21/Q2

(a) Define *momentum*.

....

.... [1]

(b) Two balls X and Y, of equal diameter but different masses 0.24 kg and 0.12 kg respectively, slide towards each other on a frictionless horizontal surface, as shown in Fig. 2.1.

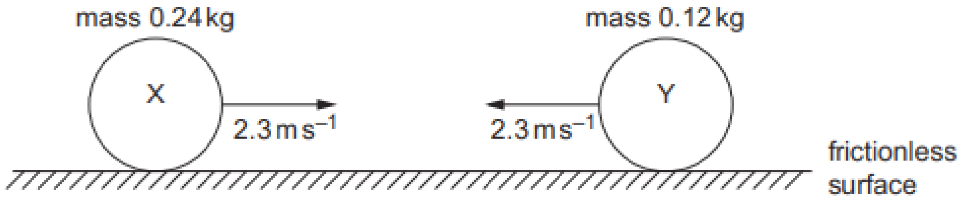


Fig. 2.1

Both balls have initial speed 2.3 m s^{-1} before they collide with each other. Fig. 2.2 shows the variation with time t of the force F_Y exerted on ball Y by ball X during the collision.

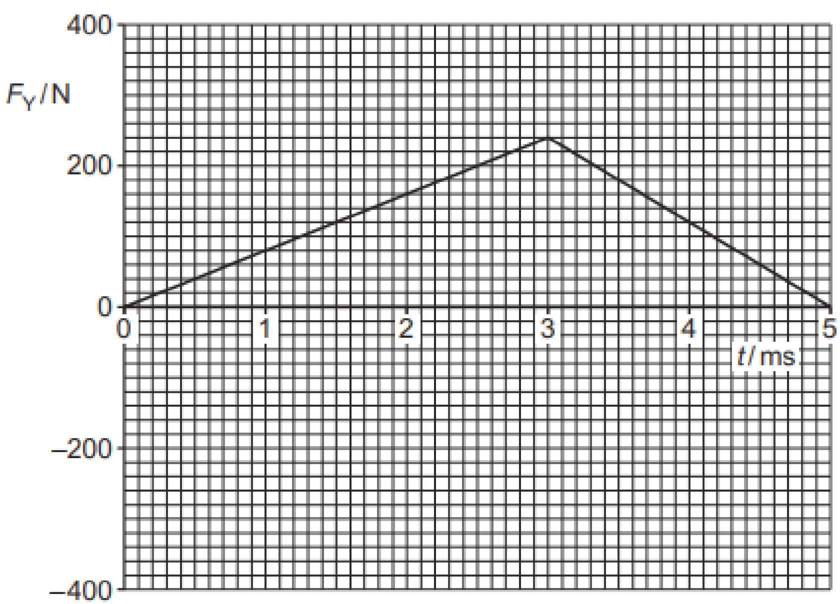


Fig. 2.2

(i) Calculate the kinetic energy of ball X before the collision.

kinetic energy = ... J [3]

(ii) The area enclosed by the lines and the time axis in Fig. 2.2 represents the change in momentum of ball Y during the collision.

Determine the magnitude of the change in momentum of ball Y.

change in momentum = ... N s [2]

(iii) Calculate the magnitude of the velocity of ball Y after the collision.

velocity = ... m s^{-1} [2]

(c) On Fig. 2.3, sketch the variation with time t of the force F_X exerted on ball X by ball Y during the collision in **(b)**.

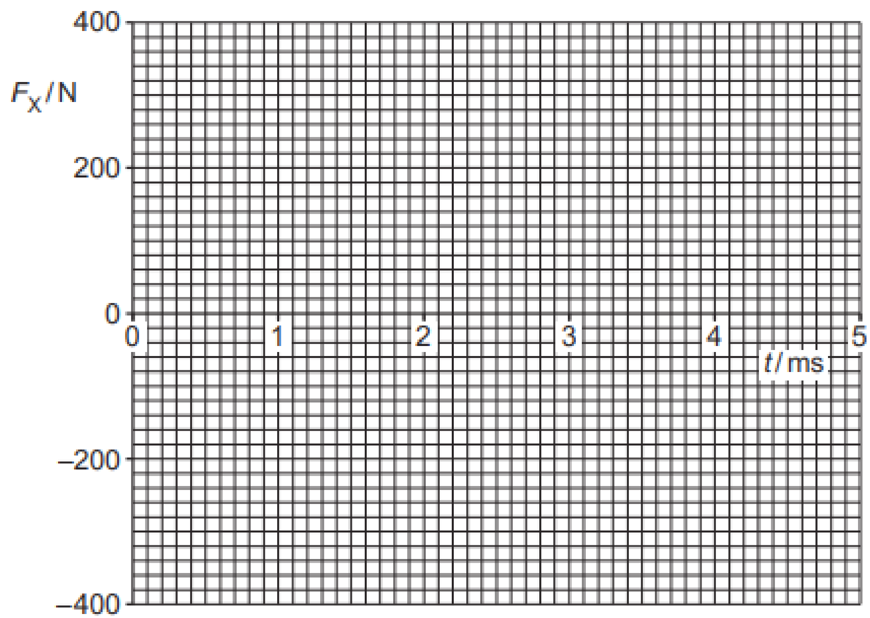


Fig. 2.3

[3]

32. 9702/22/F/M/22/Q1

A sphere of radius 2.1 mm falls with terminal (constant) velocity through a liquid, as shown in Fig. 1.1.

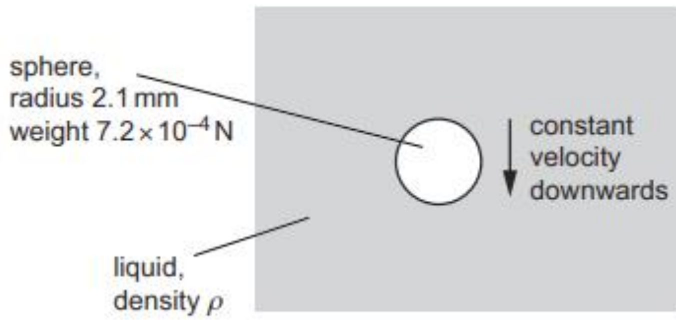


Fig. 1.1

Three forces act on the moving sphere. The weight of the sphere is 7.2×10^{-4} N and the upthrust acting on it is 4.8×10^{-4} N. The viscous force F_V acting on the sphere is given by

$$F_V = krv$$

where r is the radius of the sphere, v is its velocity and k is a constant. The value of k in SI units is 17.

(a) Determine the SI base units of k.

SI base units .... [2]

(b) Use the value of the upthrust acting on the sphere to calculate the density ρ of the liquid.

ρ = ... $kg\,m^{-3}$ [3]

(c) **(i)** On the sphere in Fig. 1.1, draw three arrows to show the directions of the weight *W*, the upthrust *U* and the viscous force F_V. Label these arrows *W*, *U* and F_V respectively. [1]

(ii) Determine the magnitude of the terminal (constant) velocity of the sphere.

velocity = ... $m\,s^{-1}$ [2]

[Total: 8]

33. 9702/22/F/M/22/Q3

A jet of water hits a vertical wall at right angles, as shown in Fig. 3.1.

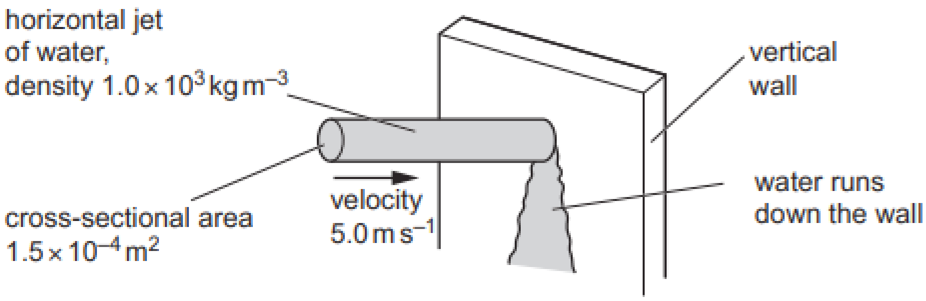


Fig. 3.1 (not to scale)

The water hits the vertical wall with a velocity of $5.0\,m\,s^{-1}$ in a horizontal direction. The cross-sectional area of the jet is $1.5 \times 10^{-4}\,m^2$. The density of the water is $1.0 \times 10^3\,kg\,m^{-3}$.

The water runs down the wall after hitting it.

(a) Show that, over a time of 1.6 s, the mass of water hitting the wall is 1.2 kg.

[2]

(b) Calculate:

(i) the decrease in the horizontal momentum of the mass of water in **(a)** due to hitting the wall

decrease in momentum = ... N s [1]

(ii) the magnitude of the horizontal force exerted on the water by the wall.

force = ... N [1]

(c) State and explain the magnitude of the horizontal force exerted on the wall by the water.

....

.... [1]

(d) Calculate the pressure exerted on the wall by the water.

pressure = ... Pa [2]

[Total: 7]

34. 9702/21/M/J/22/Q1

(a) Define velocity.

....

.... [1]

(b) A rock of mass 7.5 kg is projected vertically upwards from the surface of a planet. The rock leaves the surface of the planet with a speed of $4.0\,\text{m s}^{-1}$ at time $t = 0$. The variation with time t of the velocity v of the rock is shown in Fig. 1.1.

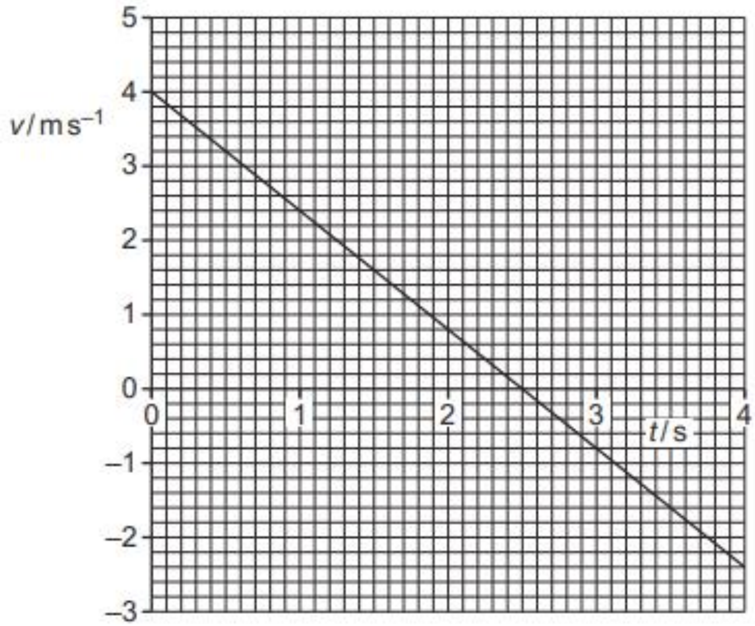


Fig. 1.1

Assume that the planet does not have an atmosphere and that the viscous force acting on the rock is always zero.

(i) Determine the height of the rock above the surface of the planet at time $t = 4.0\,\text{s}$.

height = ... m [3]

(ii) Determine the change in the momentum of the rock from time $t = 0$ to time $t = 4.0$ s.

change in momentum = ... N s [2]

(iii) Determine the weight W of the rock on this planet.

W = ... N [2]

(c) In practice, the planet in **(b)** does have an atmosphere that causes a viscous force to act on the moving rock.

State and explain the variation, if any, in the resultant force acting on the rock as it moves vertically upwards.

....

....

....

.... [2]

[Total: 10]

35. 9702/22/M/J/22/Q2

A sphere is attached by a metal wire to the horizontal surface at the bottom of a river, as shown in Fig. 2.1.

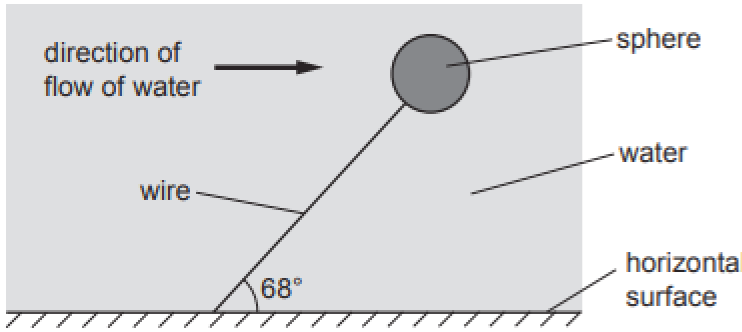


Fig. 2.1 (not to scale)

The sphere is fully submerged and in equilibrium, with the wire at an angle of 68° to the horizontal surface. The weight of the sphere is 32 N. The upthrust acting on the sphere is 280 N. The density of the water is $1.0 \times 10^3\,\text{kg}\,\text{m}^{-3}$.

Assume that the force on the sphere due to the water flow is in a horizontal direction.

(a) By considering the components of force in the vertical direction, determine the tension in the wire.

tension = ... N [2]

(b) For the sphere, calculate:

(i) the volume

volume = ... m^3 [1]

(ii) the density.

density = ... $\text{kg}\,\text{m}^{-3}$ [2]

(c) The centre of the sphere is initially at a height of 6.2 m above the horizontal surface. The speed of the water then increases, causing the sphere to move to a different position. This movement of the sphere causes its gravitational potential energy to decrease by 77 J.

Calculate the final height of the centre of the sphere above the horizontal surface.

height = ... m [3]

(d) The extension of the wire increases when the sphere changes position as described in **(c)**. The wire obeys Hooke's law.

(i) State a symbol equation that gives the relationship between the tension T in the wire and its extension x. Identify any other symbol that you use.

....

.... [1]

(ii) Before the sphere changed position, the initial elastic potential energy of the wire was 0.65 J. The change in position of the sphere causes the extension of the wire to double.

Calculate the final elastic potential energy of the wire after the sphere has changed position.

final elastic potential energy = ... J [2]

[Total: 11]

36. 9702/22/M/J/22/Q4

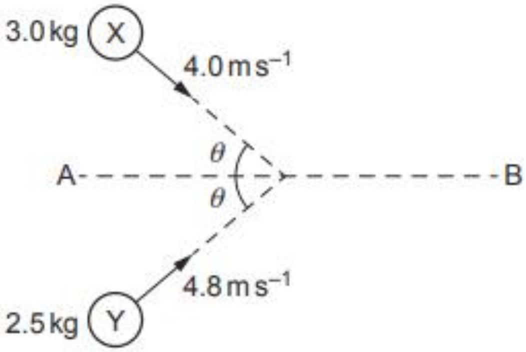
(a) State the principle of conservation of momentum.

....

....

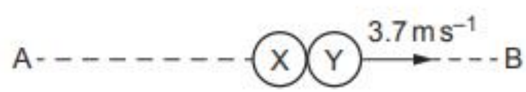
.... [2]

(b) Two balls, X and Y, move along a horizontal frictionless surface, as shown from above in Fig. 4.1.



before collision

Fig. 4.1 (not to scale)



after collision

Fig. 4.2 (not to scale)

Ball X has a mass of 3.0 kg and a velocity of $4.0\,m\,s^{-1}$ in a direction at angle θ to a line AB. Ball Y has a mass of 2.5 kg and a velocity of $4.8\,m\,s^{-1}$ in a direction at angle θ to the line AB.

The balls collide and stick together. After colliding, the balls have a velocity of $3.7\,m\,s^{-1}$ along the line AB on the horizontal surface, as shown in Fig. 4.2.

(i) By considering the components of the momenta along the line AB, calculate θ.

θ = ... ° [3]

(ii) By calculation of kinetic energies, state and explain whether the collision of the balls is inelastic or perfectly elastic.

....

.... [2]

[Total: 7]

37. 9702/23/M/J/22/Q3

(a) Define velocity.

....

.... [1]

(b) A constant driving force of 2400 N acts on a car of mass 1200 kg. The car accelerates from rest in a straight line along a horizontal road.

Assume that the resistive forces acting on the car are negligible.

(i) Calculate the acceleration of the car.

acceleration = ... $m s^{-2}$ [1]

(ii) On Fig. 3.1, sketch a graph showing the variation with time t of the velocity v of the car for the first 20 seconds of its motion.

Label this line A.

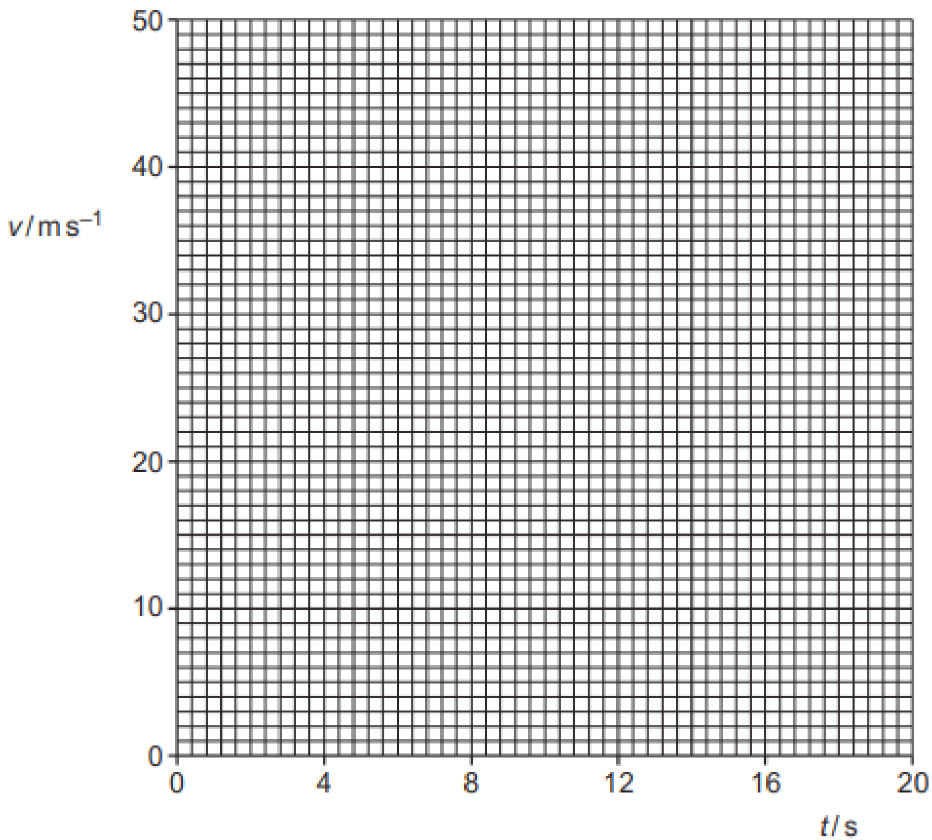


Fig. 3.1

[2]

(c) In reality, a resistive force due to air resistance acts on the car in **(b)**. This resistive force increases with speed until it becomes equal in magnitude to the driving force at time t = 12 s.

(i) On Fig. 3.1, sketch a second line to show the variation with time t of the velocity v of the car for the first 20 seconds of its motion. Label this line B. [3]

(ii) At time t = 20 s, the driving force is increased to 3000 N and remains constant at this value.

Describe how the velocity of the car changes due to this increase in the driving force.

....

....

....

.... [2]

[Total: 9]

38. 9702/22/O/N/22/Q2

A spherical balloon is filled with a fixed mass of gas. A small block is connected by a string to the balloon, as shown in Fig. 2.1.

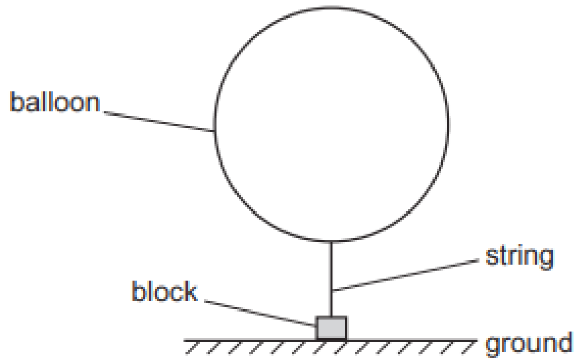


Fig. 2.1 (not to scale)

The block is held on the ground by an external force so that the string is vertical. The density of the air surrounding the balloon is $1.2\,kg\,m^{-3}$. The upthrust acting on the balloon is $0.071\,N$. The upthrust acting on the string and block is negligible.

(a) By using Archimedes' principle, calculate the radius r of the balloon.

r = ... m [2]

(b) The total weight of the balloon, string and block is $0.053\,N$.

The external force holding the block on the ground is removed so that the released block is lifted vertically upwards by the balloon.

Calculate the acceleration of the block immediately after it is released.

acceleration = ... $m\,s^{-2}$ [3]

(c) The balloon continues to lift the block. The string breaks as the block is moving vertically upwards with a speed of $1.4\,\mathrm{m\,s^{-1}}$. After the string breaks, the detached block briefly continues moving upwards before falling vertically downwards to the ground. The block hits the ground with a speed of $3.6\,\mathrm{m\,s^{-1}}$.

Assume that the air resistance on the block is negligible.

(i) By considering the motion of the block after the string breaks, calculate the height of the block above the ground when the string breaks.

height = ... m [2]

(ii) The string breaks at time $t = 0$ and the block hits the ground at time $t = T$.

On Fig. 2.2, sketch a graph to show the variation of the velocity v of the block with time t from $t = 0$ to $t = T$.

Numerical values of t are not required. Assume that v is positive in the upward direction.

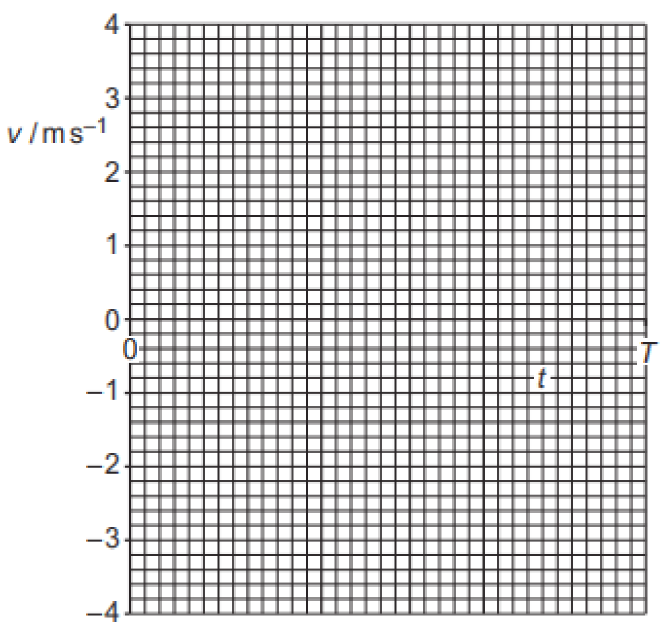


Fig. 2.2

[2]

39. 9702/23/O/N/22/Q2

The engine of a toy rocket pushes gases vertically downwards and this results in the rocket accelerating vertically upwards from the ground.

The rocket starts to move from rest at time $t = 0$. The variation with time t of the vertical velocity v of the rocket for the first 0.30 s of the flight is shown in Fig. 2.1.

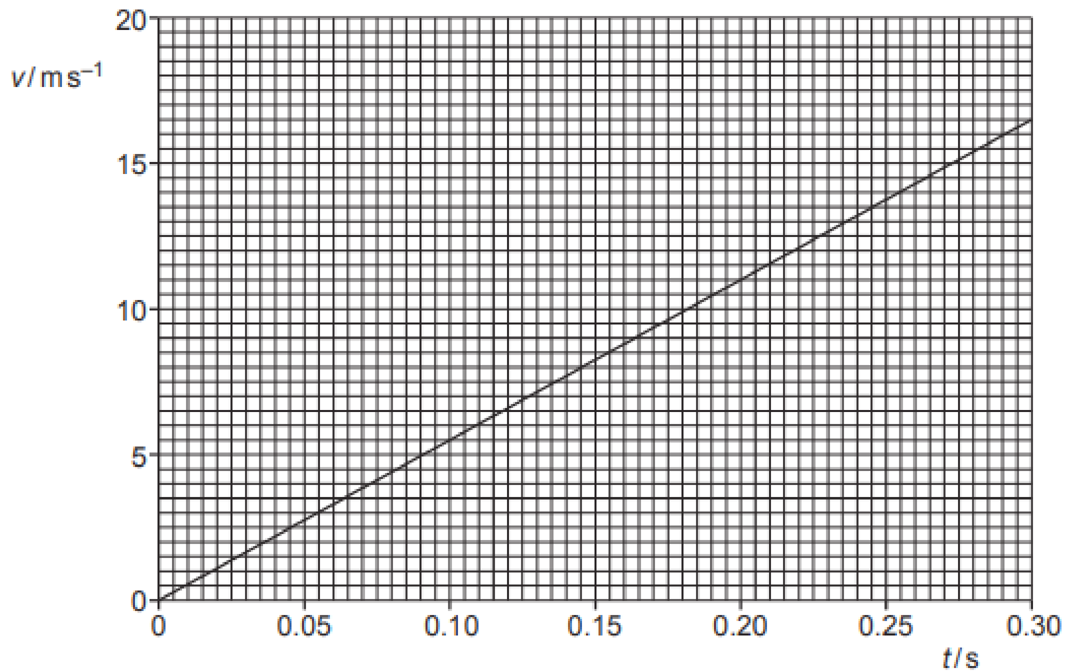


Fig. 2.1

As the rocket moves, the thrust force T provided by the rocket engine is 16 N.
Assume that the mass of the rocket is constant for this part of its flight.

Assume that air resistance is negligible.

(a) For this part of the rocket's flight:

(i) show that the acceleration of the rocket is $55\,\text{m}\,\text{s}^{-2}$

[1]

(ii) state an expression for the resultant force F experienced by the rocket in terms of the thrust force T and the weight W of the rocket

[1]

(iii) calculate the mass of the rocket.

mass = ... kg [2]

(b) At time $t = 0.30\,s$, a small piece of metal separates from the rocket.

Calculate:

(i) the height of the rocket above the ground at $t = 0.30\,s$

height = ... m [2]

(ii) the speed at which the piece of metal strikes the ground.

speed = ... $m\,s^{-1}$ [3]

[Total: 9]

40. 9702/22/F/M/23/Q4

Two blocks slide directly towards each other along a frictionless horizontal surface, as shown in Fig. 4.1. The blocks collide and then move as shown in Fig. 4.2.

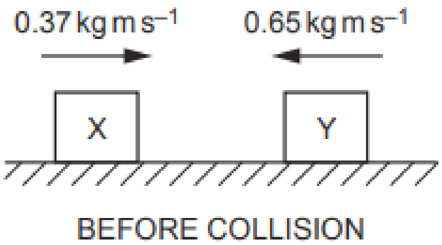


Fig. 4.1

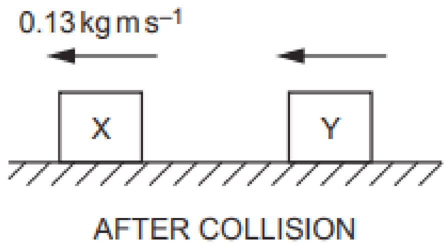


Fig. 4.2

Block X initially moves to the right with a momentum of $0.37\,kg\,m\,s^{-1}$. Block Y initially moves to the left with a momentum of $0.65\,kg\,m\,s^{-1}$. After the blocks collide, block X moves to the left back along its original path with a momentum of $0.13\,kg\,m\,s^{-1}$. Block Y also moves to the left after the collision.

(a) Block X has an initial kinetic energy of 0.30 J.

Calculate the mass of block X.

mass = ... kg [3]

(b) Determine the magnitude of the momentum of block Y after the collision.

momentum = ... $kg\,m\,s^{-1}$ [1]

(c) Block X exerts an average force of 7.7 N on block Y during the collision.

Calculate the time that the blocks are in contact with each other.

time = ... s [2]

[Total: 6]

41. 9702/21/M/J/23/Q3

A block is pulled in a straight line along a rough horizontal surface by a varying force X, as shown in Fig. 3.1.

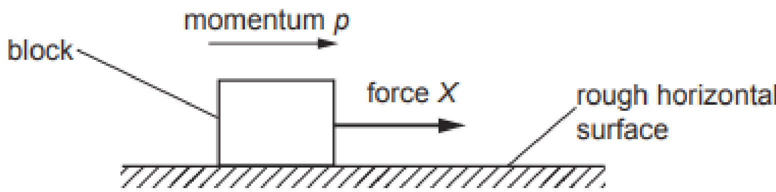


Fig. 3.1

Air resistance is negligible. Assume that the frictional force exerted on the block by the surface is constant and has magnitude 2.0 N.

The variation with time t of the momentum p of the block is shown in Fig. 3.2.

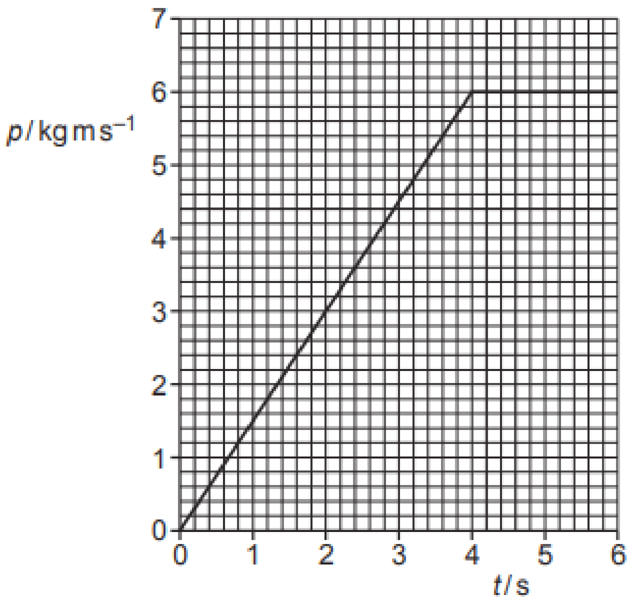


Fig. 3.2

(a) State Newton's second law of motion.

....

.... [1]

(b) Use Fig. 3.2 to determine, for the block at time t = 2.0 s, the magnitude of:

(i) the resultant force on the block

resultant force = ... N [1]

(ii) the force X.

X = ... N [1]

(c) On Fig. 3.3, sketch a graph to show the variation of force X with time t from t = 0 to t = 6.0 s.

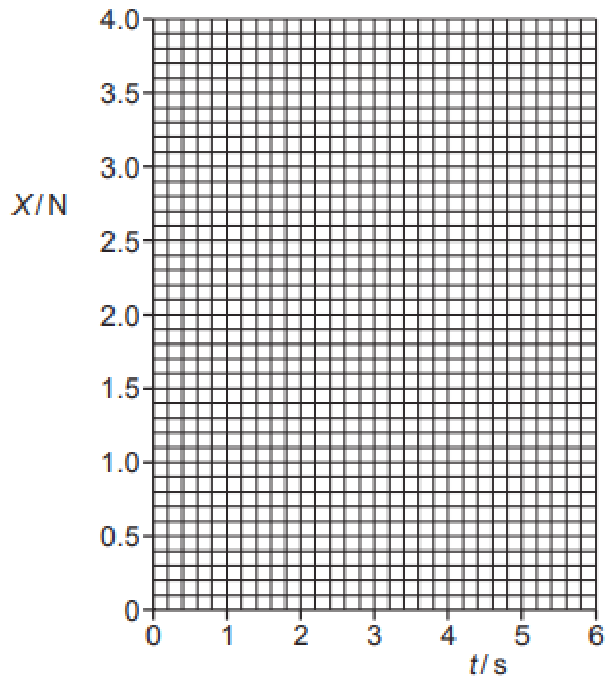


Fig. 3.3

[3]

[Total: 6]

42. 9702/22/M/J/23/Q3

A block is pulled by a force X in a straight line along a rough horizontal surface, as shown in Fig. 3.1.

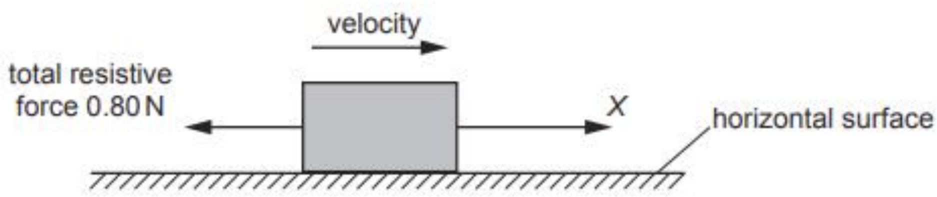


Fig. 3.1

Assume that the total resistive force opposing the motion of the block is 0.80 N at all speeds of the block.

The variation with time t of the magnitude of the force X is shown in Fig. 3.2.

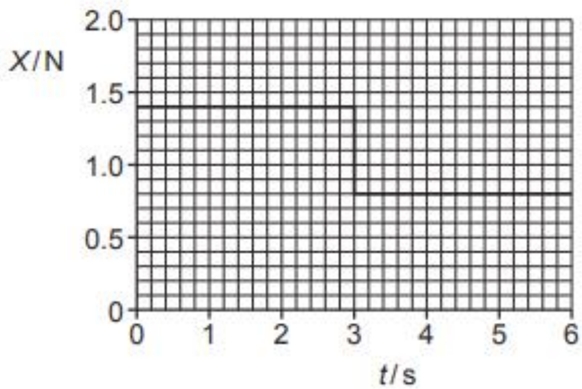


Fig. 3.2

(a) **(i)** Define force.

....

.... [1]

(ii) Determine the change in momentum of the block from time $t = 0$ to time $t = 3.0$ s.

change in momentum = ... $kg\,m\,s^{-1}$ [2]

(b) (i) Describe and explain the motion of the block between time t = 3.0 s and time t = 6.0 s.

....

....

....

.... [2]

(ii) Force X produces a total power of 2.0 W when moving the block between time t = 3.0 s and time t = 6.0 s.

Calculate the distance moved by the block during this time interval.

distance = ... m [3]

(c) The block is at rest at time t = 0.

On Fig. 3.3, sketch a graph to show the variation of the momentum of the block with time t from t = 0 to t = 6.0 s.
Numerical values of momentum are not required.

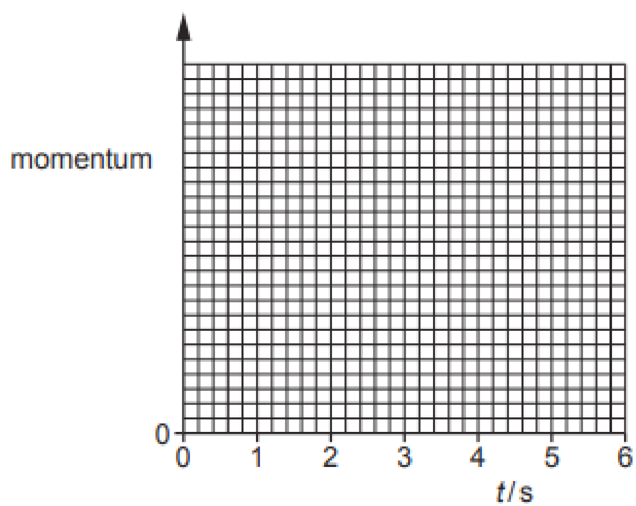


Fig. 3.3

[2]

43. 9702/23/M/J/23/Q2

A sphere floats in equilibrium on the surface of sea water of density $1050\,kg\,m^{-3}$, as shown in Fig. 2.1.

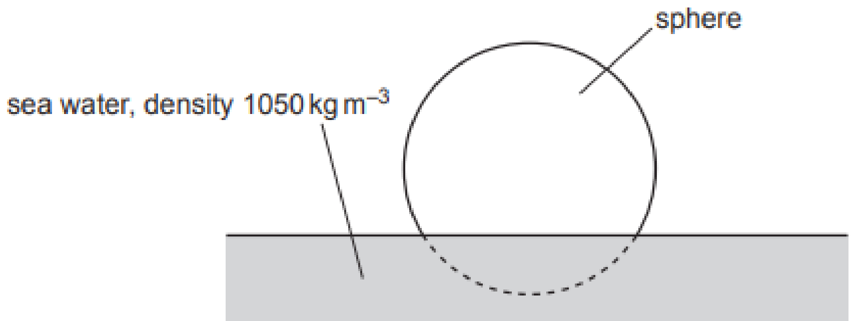


Fig. 2.1

(a) 21% of the volume of the sphere is below the surface of the water.

Calculate the density of the sphere.

density = ... $kg\,m^{-3}$ [2]

(b) The sphere is now held so that its entire volume is below the surface of the water. The sphere is then released.

(i) Calculate the initial acceleration of the sphere.

acceleration = ... $m\,s^{-2}$ [3]

(ii) The sphere accelerates upwards but remains entirely below the surface of the water.

State and explain what happens to the acceleration of the sphere as its velocity begins to increase.

....

....

....

.... [3]

[Total: 8]

44. 9702/23/M/J/23/Q3

(a) State the principle of conservation of momentum.

....

....

.... [2]

(b) A firework is initially stationary. It explodes into three fragments A, B and C that move in a horizontal plane, as shown in the view from above in Fig. 3.1.

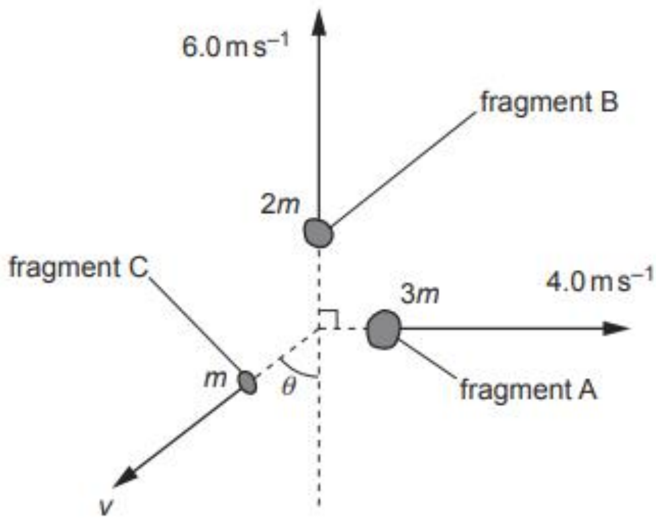


Fig. 3.1

Fragment A has a mass of $3m$ and moves away from the explosion at a speed of $4.0\,\mathrm{m\,s^{-1}}$.

Fragment B has a mass of $2m$ and moves away from the explosion at a speed of $6.0\,\mathrm{m\,s^{-1}}$ at right angles to the direction of A.

Fragment C has a mass of m and moves away from the explosion at a speed v and at an angle θ as shown in Fig. 3.1.

Calculate:

(i) the angle θ

θ = ...° [3]

(ii) the speed v.

v = ... $m\,s^{-1}$ [2]

(c) The firework in **(b)** contains a chemical that has mass 5.0 g and has chemical energy per unit mass $700\,J\,kg^{-1}$. When the firework explodes, all of the chemical energy is transferred to the kinetic energy of fragments A, B and C.

(i) Show that the total chemical energy in the firework is 3.5 J.

[1]

(ii) Calculate the mass m.

m = ... kg [3]

[Total: 11]

45. 9702/21/O/N/23/Q2

A hot-air balloon floats just above the ground. The balloon is stationary and is held in place by a vertical rope, as shown in Fig. 2.1.

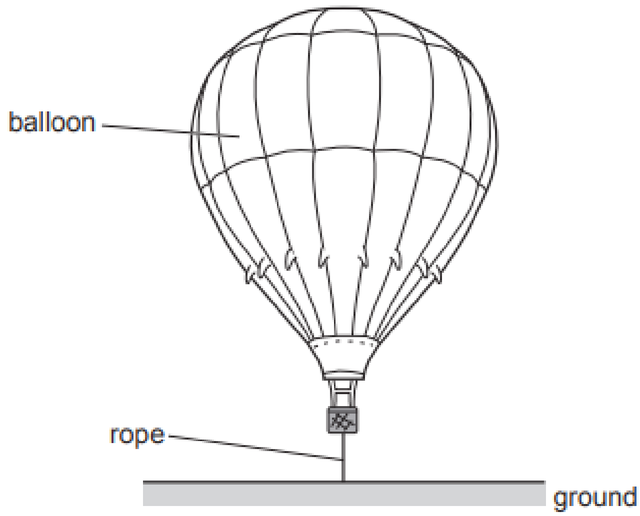


Fig. 2.1

The balloon has a weight W of 3.39×10^4 N. The tension T in the rope is 4.00×10^2 N.
Upthrust U acts on the balloon.
The density of the surrounding air is $1.23\,\text{kg}\,\text{m}^{-3}$.

(a) **(i)** On Fig. 2.1, draw labelled arrows to show the directions of the three forces acting on the balloon. [2]

(ii) Calculate the volume, to three significant figures, of the balloon.

volume = ... m^3 [3]

(iii) The balloon is released from the rope.

Calculate the initial acceleration of the balloon.

acceleration = ... $\text{m}\,\text{s}^{-2}$ [3]

(b) The balloon is stationary at a height of 500 m above the ground. A tennis ball is released from rest and falls vertically from the balloon.

A passenger in the balloon uses the equation $v^2 = u^2 + 2as$ to calculate that the ball will be travelling at a speed of approximately $100\,\mathrm{m\,s^{-1}}$ when it hits the ground.

Explain why the actual speed of the ball will be much lower than $100\,\mathrm{m\,s^{-1}}$ when it hits the ground.

....

....

....

.... [3]

(c) Before the balloon is released, the rope holding the balloon has a strain of 2.4×10^{-5}. The rope has an unstretched length of 2.5 m. The rope obeys Hooke's law.

(i) Show that the extension of the rope is 6.0×10^{-5} m.

[1]

(ii) Calculate the elastic potential energy E_P of the rope.

E_P = ... J [2]

(iii) The rope holding the balloon is replaced with a new one of the same original length and cross-sectional area. The tension is unchanged and the new rope also obeys Hooke's law.

The new rope is made from a material of a lower Young modulus.

State and explain the effect of the lower Young modulus on the elastic potential energy of the rope.

....

....

.... [2]

46. 9702/21/O/N/23/Q3

A trolley A moves along a horizontal surface at a constant velocity towards another trolley B which is moving at a lower constant speed in the same direction. Fig. 3.1 shows the trolleys at time $t = 0$.

Fig. 3.1

Table 3.1 shows data for the trolleys.

Table 3.1

trolley	mass / kg	initial speed / m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 3.2 shows the variation with time t of the velocity v for trolley B.

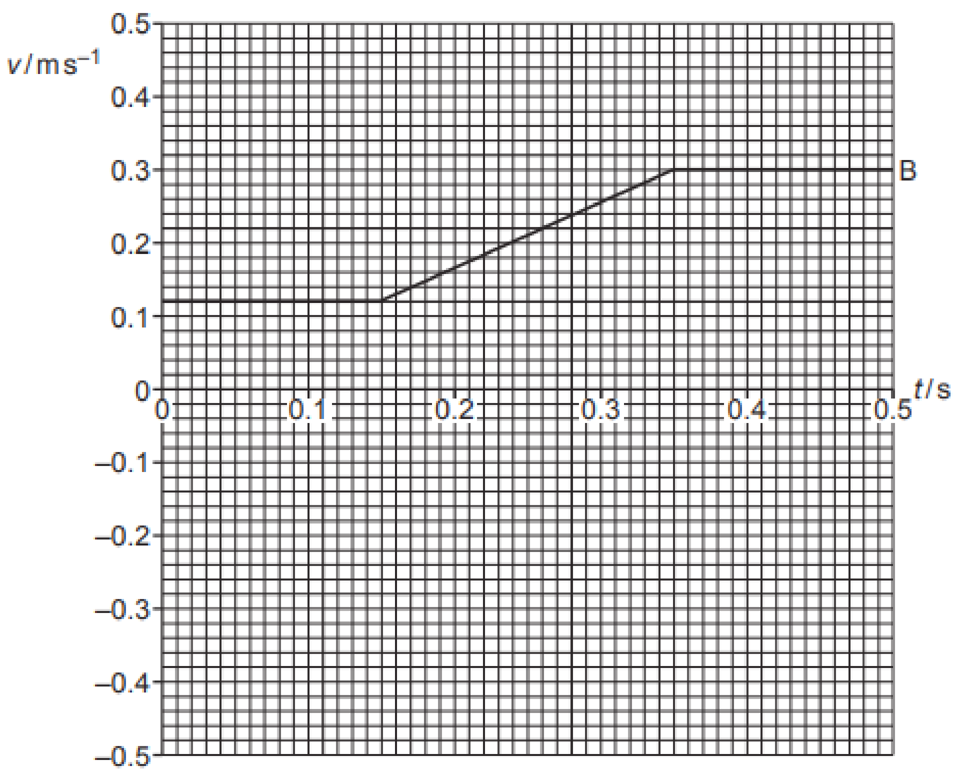


Fig. 3.2

(a) State what is represented by the area under a velocity–time graph.

.... [1]

(b) Use Table 3.1 and Fig. 3.2 to determine:

(i) the acceleration of trolley B during the collision

acceleration of B = ... $m\,s^{-2}$ [2]

(ii) the magnitude and direction of the final velocity of trolley A.

magnitude = ... $m\,s^{-1}$

direction ....

[3]

(c) On Fig. 3.2, sketch the variation of the velocity of trolley A with time t from $t = 0$ to $t = 0.50\,s$.

[3]

[Total: 9]

47. 9702/22/O/N/23/Q2

A high-altitude balloon is stationary in still air. A solid sphere is suspended from the balloon by a string, as shown in Fig. 2.1.

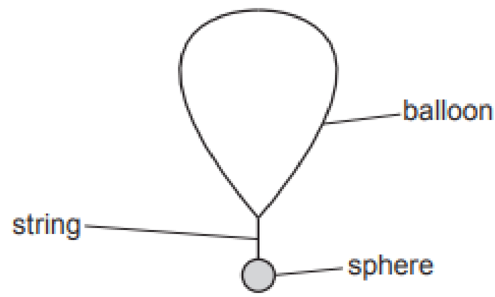


Fig. 2.1 (not to scale)

The volume of the balloon is $7.5\,m^3$. The total weight of the balloon, string and sphere is 65 N. The upthrust acting on the string and sphere is negligible.

(a) Calculate the density of the air surrounding the balloon.

density = ... $kg\,m^{-3}$ [2]

(b) The string breaks, releasing the sphere.

(i) State the magnitude of the acceleration of the sphere immediately after the string breaks.

acceleration = ... $m\,s^{-2}$ [1]

(ii) State and explain the variation, if any, in the magnitude of the acceleration of the sphere when it is moving downwards **before** it reaches terminal (constant) velocity.

....

....

....

....

....

....

.... [3]

(c) The sphere has a mass of 4.0 kg.

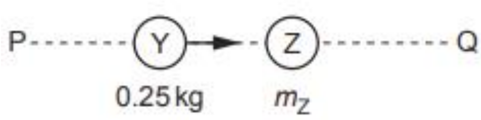
Calculate the total resistive force acting on the sphere at the instant when its acceleration is $1.9\,\text{m}\,\text{s}^{-2}$.

resistive force = ... N [2]

[Total: 8]

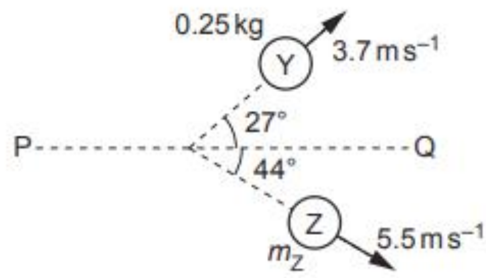
48. 9702/22/O/N/23/Q4

(a) A ball Y moves along a horizontal frictionless surface and collides with a ball Z, as illustrated in the views from above in Fig. 4.1 and Fig. 4.2.



BEFORE COLLISION

Fig. 4.1 (not to scale)



AFTER COLLISION

Fig. 4.2 (not to scale)

Ball Y has a mass of 0.25 kg and initially moves along a line PQ.
Ball Z has a mass m_Z and is initially stationary.

After the collision, ball Y has a final velocity of 3.7 m s^{-1} at an angle of 27° to line PQ and ball Z has a final velocity of 5.5 m s^{-1} at an angle of 44° to line PQ.

(i) Calculate the component of the final momentum of ball Y in the direction perpendicular to line PQ.

component of momentum = ... N s [2]

(ii) By considering the component of the final momentum of each ball in the direction perpendicular to line PQ, calculate m_Z.

m_Z = ... kg [1]

(iii) During the collision, the average force exerted on Y by Z is F_Y and the average force exerted on Z by Y is F_Z.

Compare the magnitudes and directions of F_Y and F_Z. Numerical values are not required.

magnitudes: ....

directions: ....
[2]

(b) Two blocks, A and B, move directly towards each other along a horizontal frictionless surface, as shown in the view from above in Fig. 4.3.

Fig. 4.3

The blocks collide perfectly elastically. Before the collision, block A has a speed of $4\,m\,s^{-1}$ and block B has a speed of $6\,m\,s^{-1}$. After the collision, block B moves back along its original path with a speed of $2\,m\,s^{-1}$.

Calculate the speed of block A after the collision.

speed = ... $m\,s^{-1}$ [1]

[Total: 6]

49. 9702/23/O/N/23/Q3

(a) State the conditions for a system to be in equilibrium.

....

....

.... [2]

(b) Fig. 3.1 shows an airship in flight. The airship is propelled by identical fans that can be angled to control the motion of the airship.

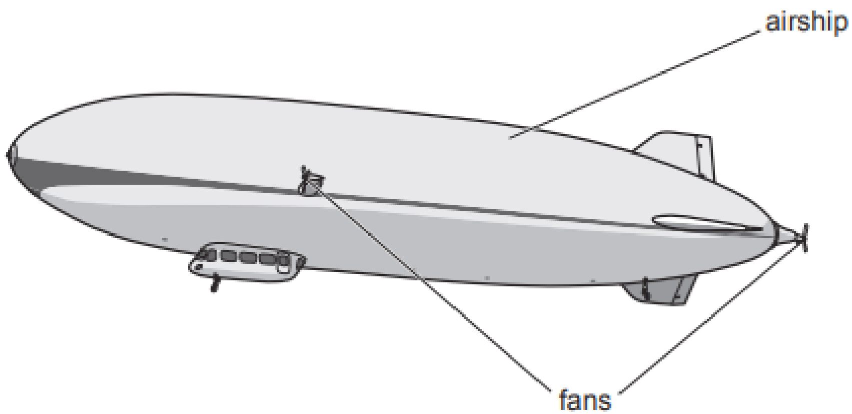


Fig. 3.1

The upthrust on the airship is 93 000 N.
The density of the surrounding air is $1.2\,kg\,m^{-3}$.

(i) Calculate the volume of air displaced by the airship.

volume = ... m^3 [1]

(ii) When fully loaded, the weight of the airship is greater than the upthrust.
To maintain horizontal flight, the fans provide a total vertical force of 3.0×10^3 N upwards on the airship.

Calculate the mass of the airship.

mass = ... kg [2]

(c) At a certain time, the airship in **(b)** is stationary. The thrust force exerted by a fan on the airship is 2800 N.

To produce this force, a mass of 64 kg of air is propelled through the blades of the fan in a time of 0.50 s. Assume that this air is initially stationary at the entrance to the fan.

Calculate:

(i) the change in momentum Δp of the air propelled through the fan blades in this time

Δp = ... $kg\,m\,s^{-1}$ [2]

(ii) the speed of the air as it leaves the fan

speed = ... $m\,s^{-1}$ [2]

(iii) the total kinetic energy of this air due to its movement through the fan.

kinetic energy = ... J [2]

[Total: 11]